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Research Article

Performance Measurements and Control Characteristics of a 1 kW Class Free Piston Stirling Converter With AC Bus Controls

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An experimental study of the control characteristics of a 1 kW class free-piston stirling converter (FPSC) with alternating current (AC) bus controls is conducted. A performance test rig is constructed using a commercial beta-type FPSC, and an AC bus controller that includes a tuning capacitor, a set of load resistors, and an AC power supply is developed. Experiment parameters include the tuning capacitance and load resistance for the AC bus circuit and the control voltage and control frequency for AC bus control under various head temperatures of the test FPSC. In addition, preliminary experiments under load resistance control scheme were conducted to identify the natural operational characteristics of the test FPSC without an external active control scheme and to determine the proper ranges of the experiment parameters for AC bus control. The experimental results are presented in terms of the output power, the power factor, and the thermal-to-electric efficiency of the FPSC with respect to the load resistance and the AC bus control parameters. In particular, the effect of the control frequency with respect to the FPSC operating frequency, as well as the influence of the alternator inductance on the control characteristics of FPSC are discussed in detail. Finally, it is anticipated that this research will yield a better physical understanding of comprehensive AC bus parameters related to the performance of FPSCs, will provide a simple and accurate control method for laboratory experiments on FPSCs, and will provide fundamental data for those who design advanced virtual tuning capacitor controllers.

Keywords: AC bus control; free-piston stirling converter; inductance; linear alternator; load resistance control

1. Introduction

1.1. Stirling Engine and Its Applications. Stirling engines have attracted attention as a key technology related to sustainable new and renewable energy due to their eco-friendliness and high efficiency rates [1–3]. A stirling engine is an external combustion engine that converts thermal energy into mechanical power based on the stirling cycle. It has no restrictions on the type of heat source and theoretically achieves the same efficiency as the Carnot cycle. Additionally, it operates with low noise and vibration due to the absence of an explosion stroke [4, 5].

Recently, free-piston stirling engines (FPSEs) are becoming more widely used due to the advantages of high durability and a compact structure. Basically, stirling engines consist of a heater, a cooler, a regenerator, a displacer piston (DP), and a power piston (PP). FPSEs have no mechanical connections between the DP and PP that causes the excessive weight and mechanical friction associated with conventional kinematic engines. Instead, the DP and PP are both connected to spring elements, forming a two-degree-of-freedom vibration system. In particular, free-piston stirling converters (FPSCs) equipped with a linear alternator have been applied in the areas of micro cogeneration, solar power generation, and micronuclear power generation [6, 7].

1.2. Control of FPSCs. FPSCs require a power control technique to adjust to environmental conditions, such as changes in heat source temperatures, load demand levels, or external disturbances. Moreover, FPSEs use vibrational piston motions

that are not constrained mechanically, meaning that vibrational instability and piston collisions can occur and in severe cases can cause an engine failure.

The power control methods of FPSCs generally include schemes which control the heat source temperature, operating pressure, and piston motion [8]. First, the heat source temperature control scheme cannot easily control the performance of FPSCs promptly because the piston motions and power changes are much higher at several tens of Hz.

The operating pressure control scheme can increase the complexity because in this case an external device is required to change the internal pressure of the FPSC, such as a compressor and valve, along with rapid and sensitive valve control actions [8].

On the other hand, piston motion control is known to be an effective and simple method to control FPSCs. It is based on the characteristic that the induced voltage in a linear alternator is proportional to the velocity of the piston/mover motion and the piston amplitude of harmonic motions as well [9].

Regarding piston motion control schemes for FPSCs, National aeronautics and space administration's (NASA's) Glenn Research Center (GRC) has developed various FPSC controllers for space exploration. These are largely classified into four types: load resistance control, Zener diode control, active power factor control, and virtual tuning capacitor control [10].

Firstly, the load resistance control and Zener diode control schemes utilize passive-type controllers. The load resistance controller controls the FPSC output voltage by adjusting the load resistance in the power generation circuit based on the characteristic that the FPSC-induced voltage is proportional to the output voltage [11]. The Zener diode controller holds the FPSC output voltage at a specific voltage using a set of Zener diodes, offering a constant breakdown voltage for reverse currents, known as the Zener voltage [12]. These methods can generate discontinuous fluctuations in the piston amplitudes upon a change in the load resistance or the Zener voltage via on/off switching. They also require a tuning capacitor for power factor correction and a rectifier circuit for electricity storage to the battery. Note that the tuning capacitor is required to compensate for the alternator inductance to ensure that the FPSC output voltage and current are in an in phase condition, thus guaranteeing a power factor of 1.

Next, active power factor control and virtual tuning capacitor control schemes are active-type controllers. Power factor correction and rectification are performed through a pulse width modulation (PWM) method without actual capacitors and rectifiers. Therefore, they have simple and compact system configurations and low power losses. Many commercial FPSCs are found to adopt these control methods [13–15], focusing on optimization of the control parameters to adjust the FPSC power output to various load demands under variable heat source conditions.

In detail, the active power factor controller matches the phase of the FPSC output current to the output voltage by adjusting the output current to a reference current which is predetermined based on the output voltage and the target power output. However, this type of controller is known to be limited to alternators with low inductance levels, and they also require the measurement of the piston amplitudes [10].

On the other hand, in virtual tuning capacitor controllers, the FPSC output voltage is forced by a control voltage that is directly applied to the alternator using an external power source. The control voltage is determined considering the tuning capacitance voltage to compensate for the alternator inductance voltage. This method requires an external power source but incurs no limitation to the alternator inductance and has no requirement to measure the piston motions. For these reasons, this approach is currently known as one of the most effective FPSC control methods [10].

1.3. Literature Review of Alternating Current (AC) Bus Control Schemes. Following the prior art of virtual tuning capacitor control, AC bus controllers have been studied with the same core principle by applying an actual tuning capacitor and supplying control voltage using an AC power supply, excluding PWM [10]. This control method is straightforward and not overly complex, making it feasible to implementation in laboratory settings. Furthermore, this method allows for precise control of piston motion, ensuring stable and reliable operation. However, when implementing a commercial digital controller, more complex technologies, such as AC-DC conversion and the generation of control voltage waveforms using PWM, are required. Several studies have focused on AC bus control methods, but studies of virtual tuning capacitors are rare.

Lewandowski and Schreiber [16] presented the basic concept of AC bus control. They revealed that the FPSC power output can be controlled by directly controlling the piston amplitude with the control voltage from experiments involving two parallel-connected FPSCs.

Gaines and Sickenberger [17] applied an AC bus control scheme to FPSCs for space exploration missions. They measured and collected performance data at various AC bus control voltages and FPSC hot end temperatures. Using the collected data, they implemented a prediction model to estimate the output voltage, current, and power factor of the FPSC.

Ko et al. [18] performed experimental studies of a 1 kW FPSC for micro cogeneration using an AC bus control technique and demonstrated the effect of the tuning capacitor on power factor correction and that of the control voltage on the FPSC power output.

Yang et al. [19] examined the impact of the tuning capacitor on the stability and power output of a fixed frequency controlled FPSC. The study discovered that raising the value of the tuning capacitor decreased the frequency and size of changes in the piston's amplitude at various temperatures. This suggests an enhancement in the operational stability of FPSC with higher capacitance. Furthermore, the tuning capacitor increases did not lead to significant power output variations. The authors emphasized optimizing the controller's capacitor to enhance FPSC stability, particularly for critical stages like space launches or high load conditions.

Recently, there has been ongoing research on FPSC controllers. Dugala et al. [20] reviewed the development and testing of the dual converter controller for NASA's Advanced Stirling Convertor systems, focusing on the controller's role in

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Table 1: Comparison of AC bus control parameter influence studies.

Ref.	AC bus control parameter				Hot end temperature influence
	Tuning capacitor	Load resistance	Control voltage	Control frequency	
Lewandowski and Schreiber [16]	Not included	Not included	Included	Not included	Not included
Gaines and Sickenberger [17]	Not included	Not included	Included	Not included	Included
Ko et al. [18]	Included	Not included	Included	Not included	Not included
Yang et al. [19]	Included	Not included	Not included	Not included	Not included

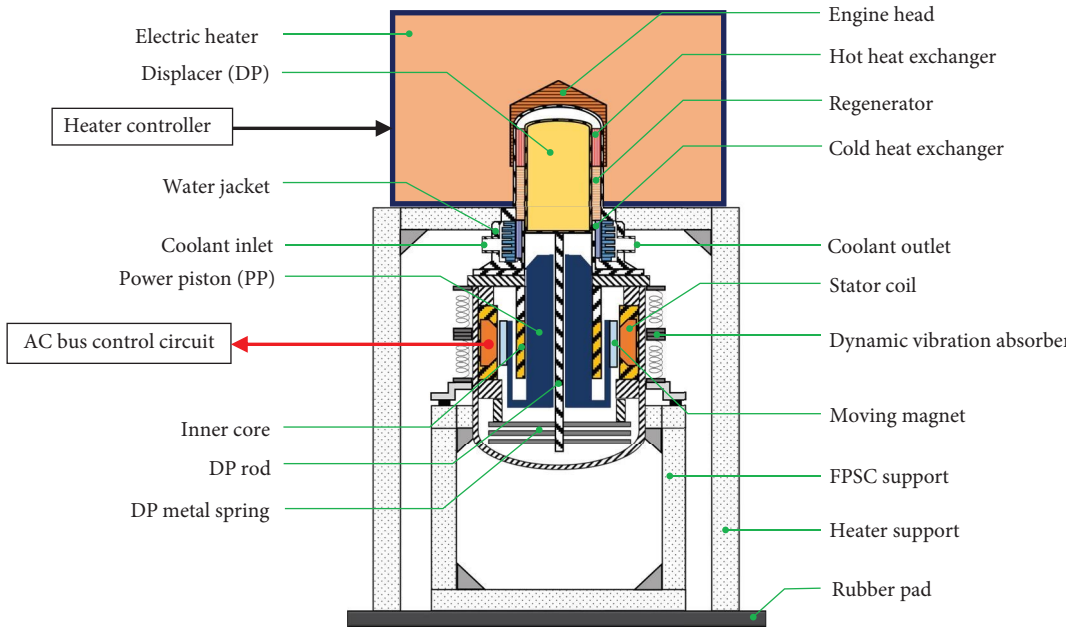


Figure 1: System configuration of the performance test rig for the test FPSC.

managing AC to DC power conversion and controlling frequency and voltage. It also highlights the controller's importance in ensuring reliable operations during space missions. Barth et al. [21] present the design of a simplified controller for a 1 kW FPSC, optimized for space applications. This study emphasizes a capacitor-based power factor correction strategy, as well as reducing complexity and increasing reliability through advanced materials. The controller aims to maximize efficiency and reliability in space, and its effectiveness was confirmed with simulations.

The primary goal of FPSC control is to ensure operational reliability by maintaining a stable power supply, even when the operating environment of the FPSC (e.g., heating/cooling temperature, load) undergoes slight variations due to disturbances. To achieve this, it is essential to investigate and understand the performance and control characteristics of the FPSC in relation to the controller's control parameters.

However, there is little experimental research offering a comprehensive and in-depth understanding of these aspects, as shown in Table 1. While existing NASA studies have explored certain relationships—such as those between control voltage and piston amplitude, and between tuning capacitors and performance—they are limited in scope and do not sufficiently address the performance characteristics of other control parameters.

1.4. Research Goals and Scope. This study is an experimental investigation into the performance measurements and control characteristics of a 1 kW class FPSC using an AC bus control scheme. We construct an FPSC performance test rig utilizing a 1 kW class commercial FPSC and a lab-implemented AC bus circuit. We identify the dynamic characteristics of the test FPSC through a series of experiments under a passive control scheme. This process allows us to identify the alternator inductance of the test FPSC and determine the appropriate tuning capacitance for power factor correction. Finally, we measure the control characteristics of the test FPSC for the AC bus control parameters, including control voltage and frequency, tuning capacitance, and load resistance of the AC bus circuit. In addition, we analyze the effects of the control frequency and the alternator inductance in detail.

2. Experimental Setup and Preliminary Tests

2.1. Test FPSC and Performance Test Rig. Figure 1 shows the configuration of the performance test rig for the test FPSC, which is largely composed of the test FPSC, an electric heater, a cooling water supply, and an AC bus control circuit.

The test FPSC is a 1 kW class commercial beta-type engine [22]. The engine head absorbs heat from the electric heater, and a water jacket rejects the heat to the cooling water. The dynamic

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Table 2: Detailed specifications of the test FPSC [22].

Parameter	Value
Maximum electrical power output	1050 W
Maximum thermal efficiency	26%
Maximum output voltage	230 V_{rms}
Operating frequency	50 Hz
Working gas	Helium
Charge pressure	30 bar
FPSC diameter	0.30 m
FPSC height	0.45 m
FPSC weight	49 kg
Noise level at 1 m	52.5 dB

absorber removes engine body vibrations generated from the test FPSC operations. The test FPSC has a maximum electric output of 1050 W, an operating frequency of 50 Hz, a maximum output voltage of 230 V_{rms}, and maximum efficiency of 26%. Detailed specifications of the test FPSC are given in Table 2.

The heat exchanger part consists of high- and low-temperature heat exchangers with an annular structure and a regenerator between them connected in series. The DP is connected to a metal spring unit with the DP rod through the PP, while the PP is supported on a gas spring of the bounce space itself without a mechanical spring, eventually forming a two-degree-of-freedom vibration system.

The linear alternator produces electric power from the PP motions with a stator coil and mover permanent magnets attached to the PP. The AC bus control circuit is connected to the stator coil, controlling the test FPSC operations and transferring the electrical output to a load resistance component.

The electric heater, a furnace type with maximum input power of 4.5 kW, raises the temperature of the inner space encompassing the engine head through natural convection. The test FPSC and electric heater are mounted on support frames separately to prevent the FPSC body vibrations from being transmitted to the electric heater. The support frames are placed on a wide rubber pad to minimize possible vibrations. Figure 2 shows an image of performance test rig for the test FPSC described above.

2.2. AC Bus Control Circuit. Figure 3 shows a conceptual schematic of the AC bus control circuit for the test FPSC, which is constructed by referring to literature from NASA's GRC [10]. The AC bus control circuit consists of the linear alternator of the test FPSC modeled with the induced voltage, coil inductance, and coil resistance, and an external circuit including a tuning capacitor, a set of load resistors, and an AC power supply acting as the AC bus. The tuning capacitor is connected in series with the linear alternator and the AC bus, while the load resistance unit is connected in parallel with the AC bus. The load resistance can be varied by switching each resistor on and off.

The AC bus control circuit including the linear alternator, tuning capacitor, and the AC bus is expressed based on Kirchhoff's voltage law as follows:

$$v_{\text{bus}} + v_{C.t} + v_{\text{emf}} + v_{R.\text{alt}} + v_{L.\text{alt}} = 0, \tag{1}$$

where v_{bus} is the control voltage from the external AC power supply, v_{emf} is the induced voltage of the linear alternator, $v_{R.\text{alt}}$ and $v_{\text{L.alt}}$ are correspondingly the electrical voltages of the resistance and inductance of the alternator coil, and $v_{C.t}$ is the tuning capacitance voltage. In addition, v_{alt} is the alternator terminal voltage and v_{out} is the output voltage corrected by the tuning capacitor, with the magnitude identical to that of v_{bus} but in the opposite phase. Note that the load resistance voltage v_{load} is determined from $v_{\text{bus}} + v_{\text{load}} = 0$.

The tuning capacitor is designed to have a power factor of 1 by canceling $v_{C.t}$ and $v_{\text{L.alt}}$ in the opposite phase, which is equivalent to impedance matching between the inductance and capacitance in a series resistor-inductor-capacitor (RLC) circuit [23]. Physically, the resonant frequencies of the FPSC and the AC bus circuit become identical.

As a result, the AC bus control voltage becomes nearly identical to the induced voltage of the linear alternator, as the coil resistance voltage is very low in general. Note that the induced voltage in a generator is defined as $V_{\text{emf}} = 2\pi f K_e X_p$, where K_e is the motor constant and f and X_p are the operating frequency and amplitude of the mover, respectively. Therefore, the piston amplitude and thus the FPSC power output can be controlled in proportion to the AC bus control voltage, based on Equation (2) [24], assuming that the motor constant and the FPSC operating frequency are constant. The motor constant of a linear alternator is generally reduced by less than 10% at the end of the design amplitude, meaning that it can be assumed to be constant within the design amplitude range [25].

$$V_{\text{bus}} \cong V_{\text{emf}} = 2\pi f K_e X_p \tag{2}$$

Regarding the power in the circuit, the load power output (dissipation) is identical to the sum of the FPSC power output and the AC bus power output from the power balance. The FPSC power output is calculated as the product of the root mean square (RMS) voltage V, the RMS current I and the power factor $\cos\varphi$, where φ is the phase angle between the voltage and current, leading to $V_{\text{alt}} I_{\text{out}} \cos\varphi_{\text{pre}}$ at the pre-capacitor and $V_{\text{out}} I_{\text{out}} \cos\varphi_{\text{post}}$ at the post-capacitor, as described in Figure 3. Note that the P_{elec} values at both the pre- and post-capacitor are identical because the capacitors are not energy-dissipative.

The AC bus power output P_{bus} is calculated in the same way using the equation $V_{\text{bus}} I_{\text{bus}} \cos\varphi_{\text{bus}}$. The load power output P_{load} is identical to $V_{\text{load}} I_{\text{load}}$, where the power factor is 1 because the voltage and current are in phase [26]. Table 3 summarizes the calculation formulas for the power outputs in the AC bus control circuit.

Figure 4 shows an image of the AC bus control system. The AC power supply (AST3001, California Instruments) can supply variable magnitudes and frequencies of the input voltage with resolutions of 0.01 V and 0.01 Hz, respectively. The tuning capacitor is an aluminum electrolytic capacitor capable of operating at up to 450 V. The load resistance, an air-cooled load

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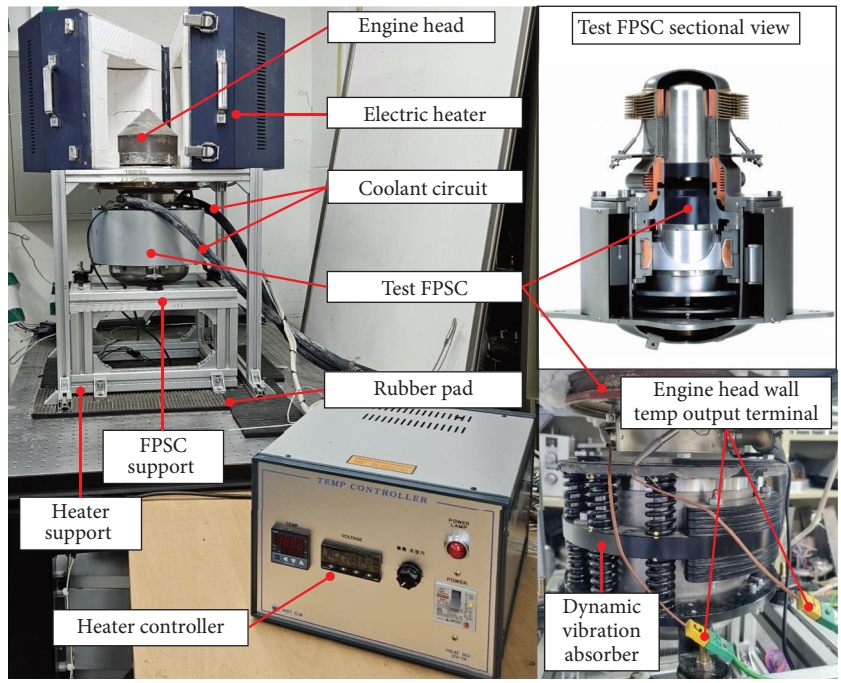


FIGURE 2: Images of the performance test rig for the test FPSC (1 kW commercial FPSC [22]).

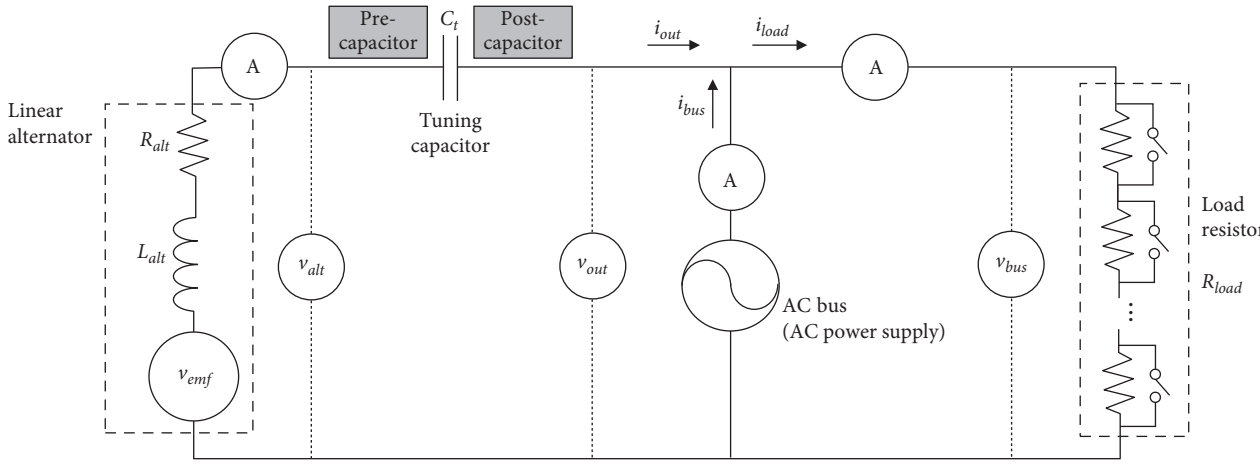


FIGURE 3: Conceptual schematic of the AC bus control circuit for the test FPSC [10].

TABLE 3: Calculation formulas for power outputs in the AC bus control circuit.

Power output type	Equations
Power balance	$P_{load} = P_{elec} + P_{bus}$
FPSC power output	$P_{elec} = V_{alt} I_{out} \cos \varphi_{pre} = V_{out} I_{out} \cos \varphi_{post}$
AC bus power output	$P_{bus} = V_{bus} I_{bus} \cos \varphi_{bus}$
Load power output	$P_{load} = V_{load} I_{load}$

bank, consists of eight 6 Ω resistors and two 2 Ω resistors. On/off switches are connected in parallel to each load resistance such that the magnitude of the load resistance can be adjusted in real time.

Figure 5 shows the measurement sensors and data acquisition process for the FPSC performance test rig with the AC bus control circuit. Four K-type thermocouples measure the temperature of the inner space of the heater (T_{heat}), the wall temperature of the engine head (T_{head}), and the inlet and outlet temperatures of the cooling water ($T_{c.in}$ and $T_{c.out}$), with data acquired through an NI module (NI9213).

A power meter (WT500, Yokogawa) measures the electric power along with the voltage and current of the FPSC power output and heater input power. An NI module measures the voltage (NI9225) and current (NI9247) at the points of the AC bus circuit specified in Figure 3. All measured data are processed using LabVIEW.

The uncertainties of the measurements consider sensor measurement errors for direct parameters (temperature,

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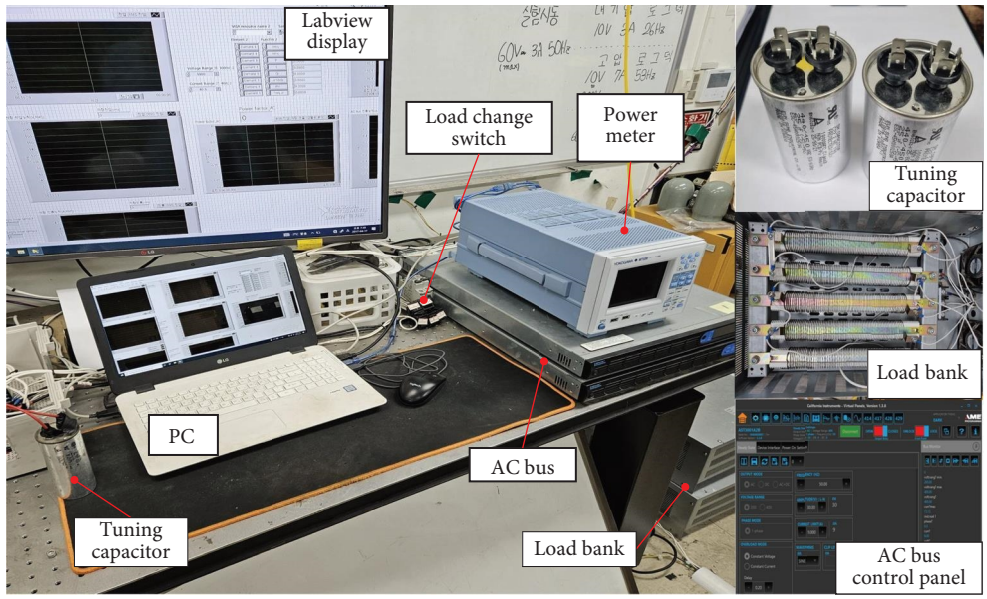


FIGURE 4: Images of the AC bus control system.

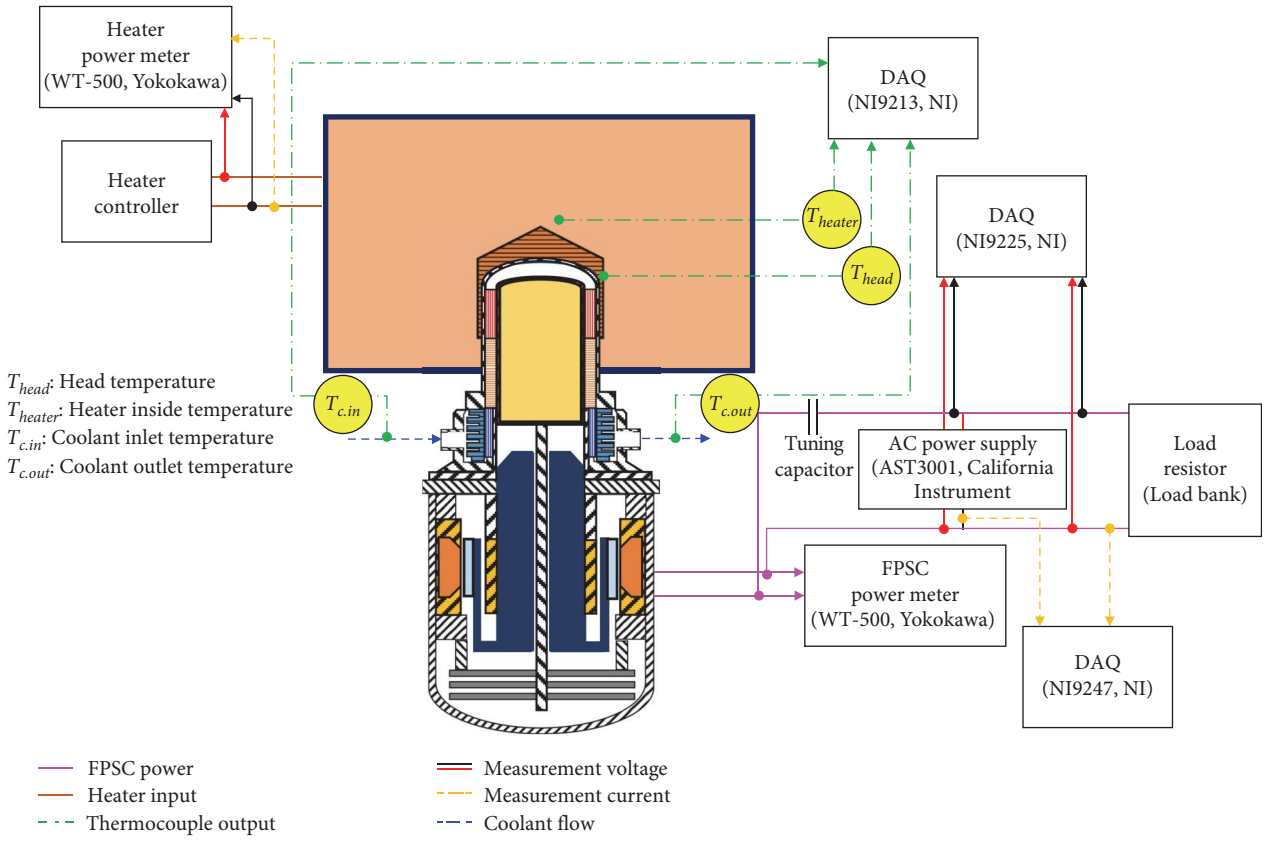


FIGURE 5: Measurement sensors and data acquisition process for the FPSC performance test rig with the AC bus control circuit.

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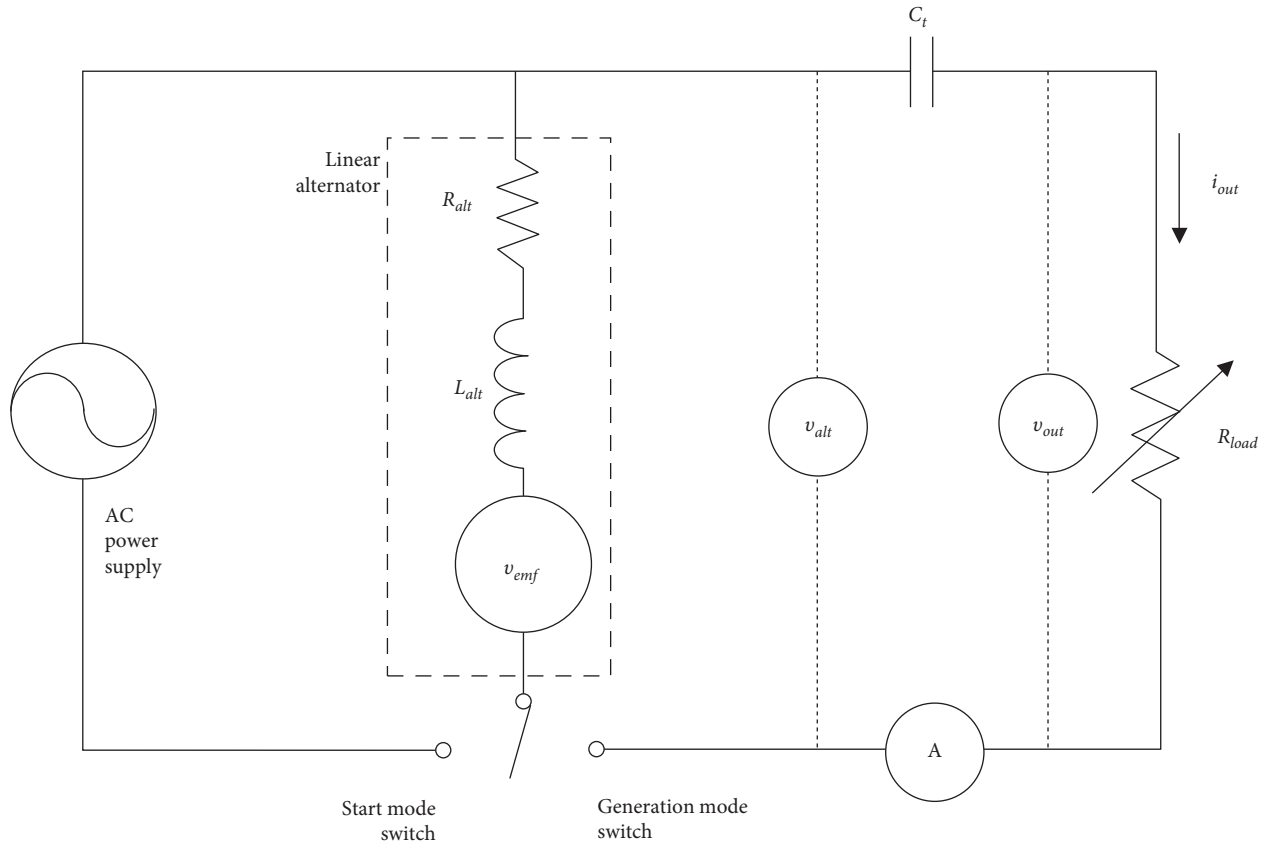


FIGURE 6: Schematic of the load resistance control circuit for the test FPSC.

voltage, and current) and their effects on indirect parameters (FPSC power output and efficiency). The uncertainties of the indirect parameters are calculated with the root sum square method, expressed as $y = \sqrt{\sum_{i=1}^{N} \left(\frac{\partial y}{\partial x_i} \delta x_i\right)^2}$, where y and δy are the indirect parameter and the corresponding uncertainty, respectively. In addition, x_i and δx_i are correspondingly the direct parameter and associated uncertainty, and N is the number of direct parameters required to calculate the indirect parameter. In this study, the sensor measurement errors from the specifications are as follows: thermocouples $\pm 0.75\%$, NI voltmeter $\pm 0.07\%$, NI ammeter $\pm 0.90\%$, and power meter $\pm 0.20\%$.

2.3. Preliminary Tests Under Load Resistance Control. In addition to the AC bus control experiments, the inherent operating characteristics of the test FPSC were measured under load resistance control, known as a basic passive control method for FPSCs without any external active control scheme [27]. Furthermore, suitable ranges of the AC bus control parameters were determined.

Figure 6 shows a schematic of the load resistance control circuit for the test FPSC, which consists of a linear alternator for the test FPSC, an AC power supply, a variable load resistance unit, a tuning capacitor, and two start/generation mode switches.

First, the operation of the linear alternator is switched from starting to generating using the on/off switch. The start mode switch connects the alternator to the AC power supply for engine startup by powering the PP, while the generation mode switch connects the alternator to the variable load resistor to produce electric power. The load resistance is variable to control the FPSC output voltage V_{alt}.

The FPSC power output P_{elec} is identical to the power dissipated in the load resistance unit. This is determined by the multiplication of the output voltage V_{out} and current I_{out}, referred to as the load voltage and current in this section. Note that the power factor of the voltage and current though the resistance is 1.

The tuning capacitance is also included in the circuit for stable FPSC operation and to correct the power factor [28]. First, the alternator inductance was measured and found to be 220.91 mH using an RLC meter with the FPSC stopped. The tuning capacitance was calculated as 45.8 μF from the impedance matching equation for RLC series circuits of $C_t = 1/(2\pi f)^2 L_{\text{alt}}$ at the FPSC specification frequency of 50 Hz [29]. Here, the tuning capacitance is selected with a standard product of 50 μF, which has a circuit resonant frequency of 49.59 Hz with error of less than 0.5 Hz from the FPSC frequency. In addition, the measured alternator resistance is found to be 4.3 Ω using an LCR meter.

Note that the alternator inductance can change with the FPSC operating conditions and performance [23]; therefore, the optimality of the tuning capacitance must be verified through FPSC operations, which is covered in the section on the AC bus control experiments.

Figure 7 shows the performance measurements of the test FPSC for variable load resistances at a head temperature of

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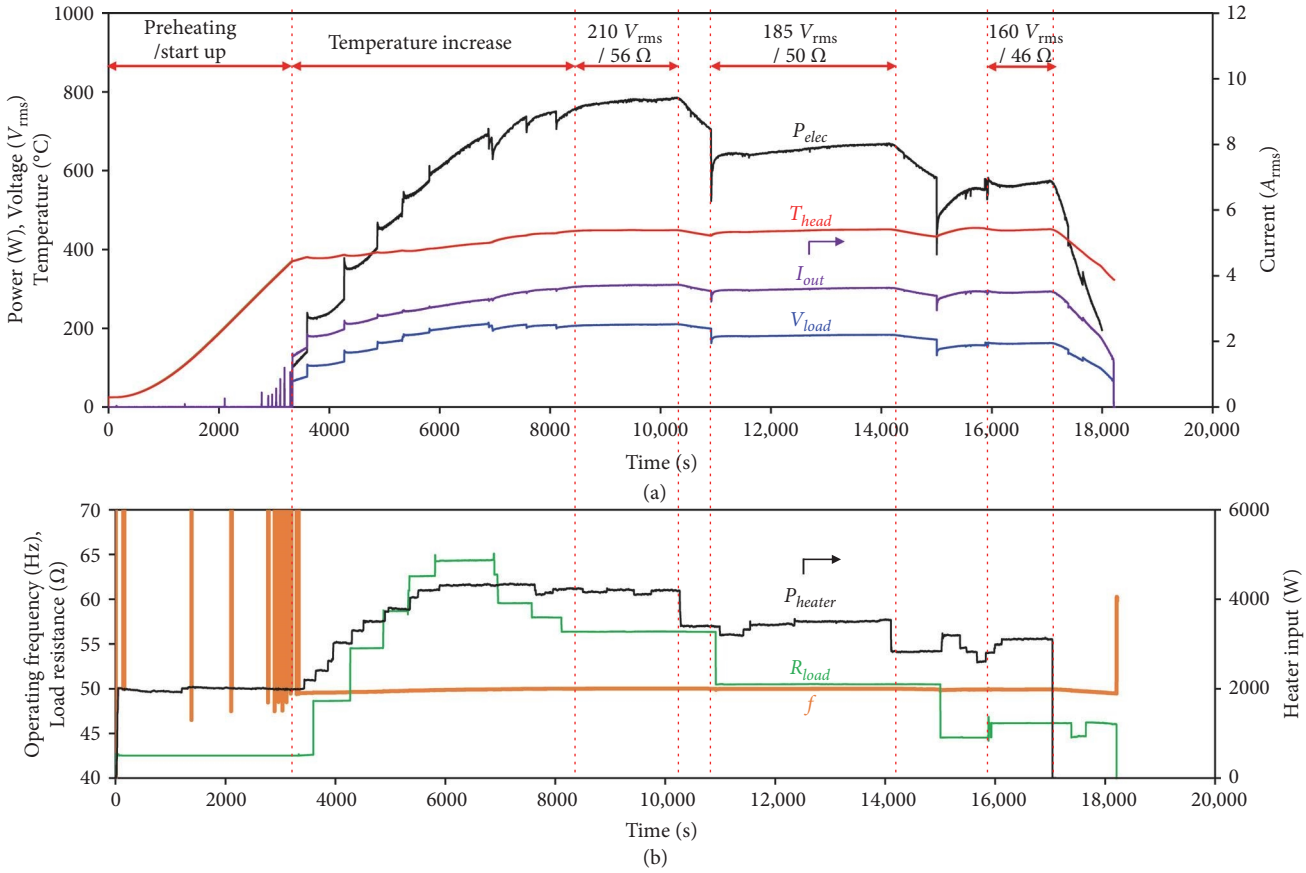


FIGURE 7: Performance measurements of the test FPSC for variable load resistances at a head temperature of 450°C: (a) FPSC power output, output voltage and current, and head temperature, and (b) heater input power, operating frequency, and load resistance.

450°C, including (a) the FPSC power output, load voltage and current, and head temperature, and (b) the heater input power, operating frequency, and load resistance.

The experiment proceeds in the order of preheating/starting FPSC (0–3350 s), increasing the head temperatures and FPSC power output (3350–8400 s), changing the load resistances (8400–17,200 s), and stopping (17,200–18,300 s).

The FPSC is preheated to an operable temperature, and by powering the PP/mover with the start switch on, it starts at a head temperature of approximately 300°C with an initial load voltage of 68 V_{rms} and operating frequency of 49.5 Hz under an initial load resistance of 42 Ω.

The head temperature is then increased to 450°C with heater power of 4300 W, and the FPSC power output reaches 785 W. The load voltage is raised to 210 V_{rms} by increasing the load resistance stepwise from 42 to 64 Ω. Note that the maximum load voltage from the specification is 230 V_{rms}.

The load voltage was controlled to the experimental points of 210, 185, and 160 V_{rms} by changing the load resistance to 56 Ω, 50 Ω, and 46 Ω, respectively, while the head temperature was maintained at 450°C by adjusting the heater input power. The operating frequency increases slightly to 50.02 Hz from the starting frequency, implying that an increase in the FPSC internal pressure under a higher head temperature leads to an increase in the pressure stiffness of the FPSC vibration system [30].

TABLE 4: Measured power output and output voltage along with the operating frequency of test FPSC versus the load resistance of 46, 50, and 56 Ω at a head temperature of 450°C.

Load resistance	46 Ω	50 Ω	56 Ω
FPSC power output (W)	582	666	785
Output voltage (V_{rms})	164.4	183.4	210.4
Operating frequency (Hz)	49.94	50.00	50.02

Note that changing the load resistance in a stepwise manner led to abrupt fluctuations in the FPSC power output, which also implies that the piston motions can be unstable, especially when the electric load is released with higher resistance levels. The FPSC was gradually stopped at a head temperature of 320°C by turning off the heater.

Table 4 summarizes the measured power output and output voltage as well as the operating frequency of the test FPSC versus the load resistances of 46, 50, and 56 Ω at the head temperature of 450°C. When the load resistance increases, which means a decrease in the electric load damping to the FPSC, the load voltage and power output increase. The piston amplitude is also estimated to increase due to the proportionality of the piston amplitude to the load voltage from Equation (2), accompanying slight increases in the operating frequency. Note that lower damping leads to increases in the

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Table 5: Experimental conditions of the AC bus control parameters at various head temperatures for the test FPSC.

AC bus parameter	Reference value	Range	Interval	Head temperature
Tuning capacitance (μF)	50	37.5–75	12.5	350–550°C (50°C interval)
Load resistance (Ω)	42	38–48	2	
Control voltage (V_{rms})	185	160–210	12.5	350–500°C (50°C interval)
Control frequency (Hz)	50	49.2–50.6	0.12	350, 450°C

resonant amplitude and damped natural frequency of mechanical vibration systems [31].

The uncertainties of the measurements for the 56 Ω case are estimated as follows: for the head temperature, the uncertainty is 3.37°C at 449°C; for the output voltage, it is 0.15 V_{rms} at 210.4 V_{rms}; for the output current, it is 0.034 A_{rms} at 3.74 A_{rms}; and for the FPSC power output, it is 31 W at 784 W.

3. Experimental Results

Experimental measurements of the FPSC performance and control characteristics are taken for the AC bus parameters, i.e., the control voltage and frequency of the AC bus control, and the tuning capacitance and load resistance of the AC bus circuit.

Table 5 summarizes the experimental conditions of the AC bus parameters at various head temperatures for the test FPSC. The reference conditions of the AC bus parameters are a tuning capacitance value of 50 μF, load resistance of 42 Ω, control voltage of 185 V_{rms}, and control frequency of 50 Hz. Each AC bus parameter is tested at multiple head temperatures between 350 and 550°C. Specific parameter ranges are detailed in the table. Note that the experimental conditions were determined by referring to the FPSC specifications and to load resistance control experiments.

The steady state condition is defined as a state in which the head temperature is maintained within ±0.5% of the target temperature for more than 10 min, and the average value for 10 min is used as a representative value. The coolant flow rate and inlet temperature were maintained at 16.3 L/min and at approximately 25°C, respectively. Note that the other AC bus parameters, except for the experimental parameter, are set to the reference values in Table 5.

3.1. Startup and Basic Operation of the Test FPSC With the AC Bus Control Scheme. Figure 8 shows the measured startup and power generation process of the test FPSC with AC bus control, where the following are indicated: (a) heater input power, head temperature, and AC bus voltage, and (b) FPSC power output, AC bus power output, and load power output. The AC bus control voltage and frequency are 185 V_{rms} and 50 Hz, respectively.

The FPSC startup sequence under the AC bus control scheme is as follows: first, the head temperature is preheated by supplying heater input power of 4000 W up to 225 s. When the head temperature reaches 43°C, AC bus voltage is applied gradually up to 185 V_{rms} up to 335 s, with the AC bus power supplied for FPSC motoring and for the load resistance.

The load power output is constant at 797 W with fixed values of the load resistance and control voltage, while the AC bus power output is as high as 1160 W and the FPSC motoring power is up to 363 W. Note that the sum of the AC bus power output and FPSC power output is identical to the load power output in the parallel AC bus circuit.

As the head temperature continues to rise, the motoring power to the FPSC decreases continuously, after which the FPSC begins to produce power output when the head temperature reaches 165°C, with all of the AC bus power then supplied to the load power output. This shows a natural transition from the motoring mode of the FPSC to the generating mode without any external switching, in contrast to earlier load resistance control schemes.

As the head temperature increases further to 500°C, the FPSC power output increases continuously to 772 W and the AC bus power output decreases to 25 W; both the FPSC power and AC bus power contribute to the load power output of 797 W. The heater input power is 4000 W to maintain the head temperature at 500°C.

The motoring power supplied from the AC bus to the test FPSC during the startup process is considered as a power loss at the system level. Figure 9 shows the FPSC power output during startup for initial head temperatures of 40, 80, and 120°C. Negative FPSC power means power input from the AC bus. The heater input power is maintained at 4000 W. The AC bus control voltage and frequency are 185 V_{rms} and 50 Hz, respectively.

As a result, the higher the preheating temperature, the lower the motoring power of the FPSC and the shorter the motoring time until power generation. Interestingly, the head temperature for power generation was constant at the head temperature of 165°C regardless of the preheating temperature.

Table 6 shows the measured motoring power and time for the test FPSC at the initial head temperatures of 40, 80, and 120°C. All motoring times, average motoring power levels, and total amounts of motoring power decrease at higher preheating temperatures, with these being reduced correspondingly to approximately half, one quarter, and one eighth at the initial head temperature of 120°C, compared to the corresponding outcomes at 40°C.

3.2. Optimal Tuning Capacitance of the AC Bus Circuit. Figure 10 shows the measured transient FPSC power output and power factor at the pre- and post-capacitors for the test FPSC under the AC bus control scheme with tuning capacitances of (a) 50 μF and (b) 75 μF. PF_{pre} and PF_{post} denote the power factor at the pre- and the post-capacitor, respectively, as given in Figure 3. The head temperature was increased to 550°C. The control voltage and frequency are 185 V_{rms} and 50 Hz, respectively. The load resistance is 42 Ω.

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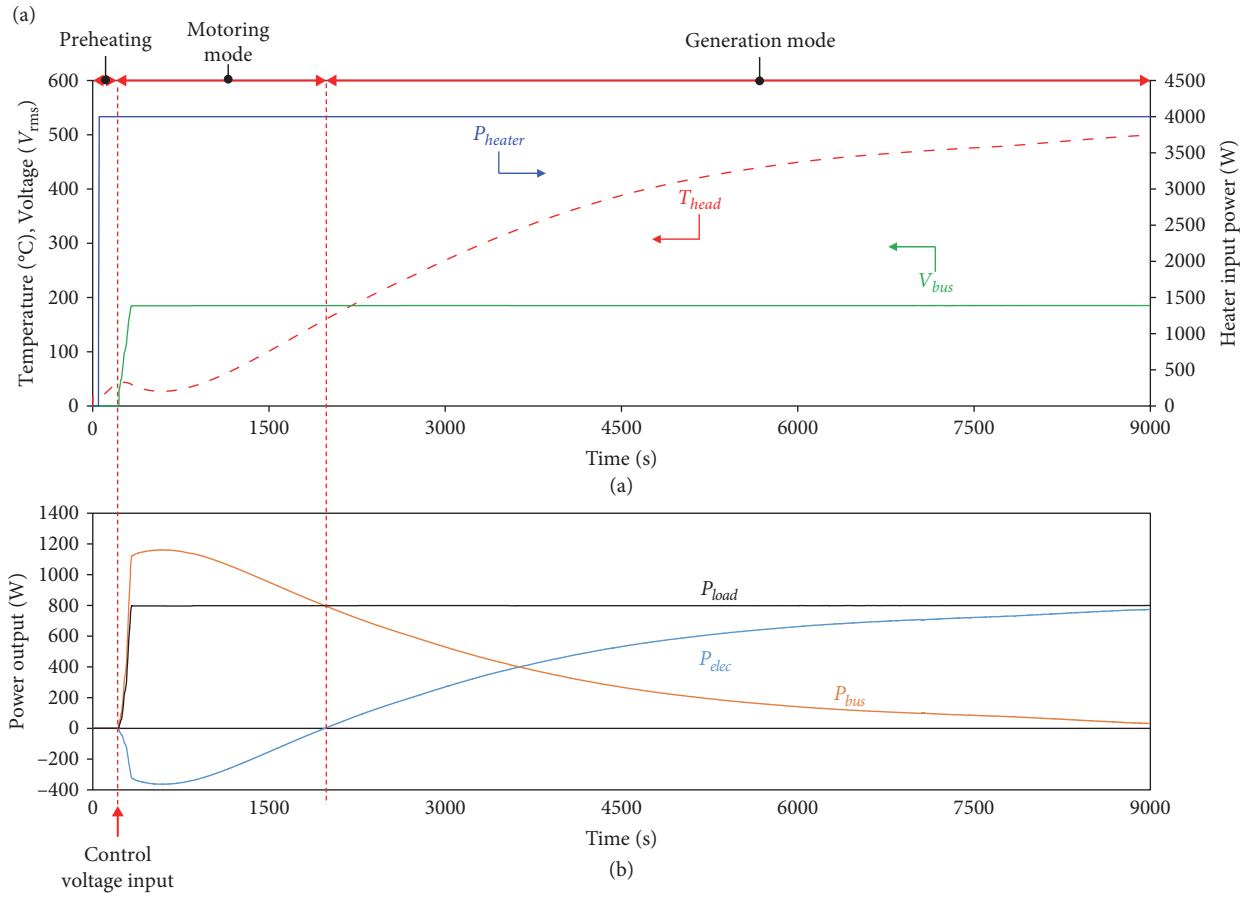


FIGURE 8: Measured startup and power generation process of the test FPSC with AC bus control: (a) heater input power, head temperature, and control voltage, and (b) FPSC power output, AC bus power output, and load power output.

At the tuning capacitance of 50 μF, the FPSC power output increased from 350 to 900 W as the head temperature increased to 550°C. Meanwhile, PF_{pre} decreased from approximately 1 to about 0.5 while PF_{post} increased from 0.9 to 0.9989. On the other hand, the experiment at 75 μF showed much lower FPSC power output of 700 W due to the lower PF_{post} of 0.8599 at the same head temperature.

Figure 11 shows the current and voltage waveforms of (a) the pre- and (b) the post-capacitor at 22,750 s in Figure 10a. The pre-capacitor waveform shows that the phase difference between the current and voltage is 60.2° with a low power factor of 0.4975. However, at the post-capacitor, the phase difference is 2.7° with a power factor of 0.9989, as the phase delay of the current by the alternator inductance is eliminated by the effect of the tuning capacitor.

Note in addition that when the engine's natural operating frequency and the electric circuit's resonant frequency match, the current–voltage phase difference becomes 0, the power factor becomes 1, and the maximum power arises [32].

Figure 12 shows the measured (a) FPSC power output, (b) AC bus power output, (c) load power output, (d) FPSC efficiency, and (e) power factor (post-capacitor) versus the tuning capacitance of 37.5, 50, 62.5, and 75 μF for head temperatures of 350, 400, 450, 500, and 550°C. Detailed experimental results for 37.5 and 62.5 μF are omitted for brevity.

The FPSC power output is maximized at the tuning capacitance of 50 μF, especially at higher head temperatures. On the other hand, the AC bus power output is lowest at that capacitance value, opposite to the trend of the FPSC power output. Note that the sum of the FPSC power output and the AC bus power output equals the load power output from the power balance. Moreover, the load power output is constant at 814.9 W regardless of the tuning capacitances and/or head temperatures due to the fixed control voltage of 185 V_{rms} relative to the fixed load resistance of 42 Ω.

The FPSC efficiency showed a trend similar to that of the FPSC power, peaking at 50 μF and increasing at higher head temperatures. Note that the FPSC efficiency is calculated from $\eta = P_{elec}/P_{heater}\eta_{heater}$, where the heater efficiency η_{heater} is 82.3%. Details pertaining to the calculation of the heater efficiency are given in the Appendix.

The power factors are maximized at the capacitance of 50 μF with a value around 1 and decrease at higher capacitances, especially as the head temperature increases. These trends are also found in the FPSC power output. As a result, it was experimentally confirmed that the FPSC power output and efficiency are highest at the tuning capacitance of 50 μF due to the higher power factors. Also note that the tuning capacitance is not effective at lower head temperatures; this finding will be covered in the discussion section.

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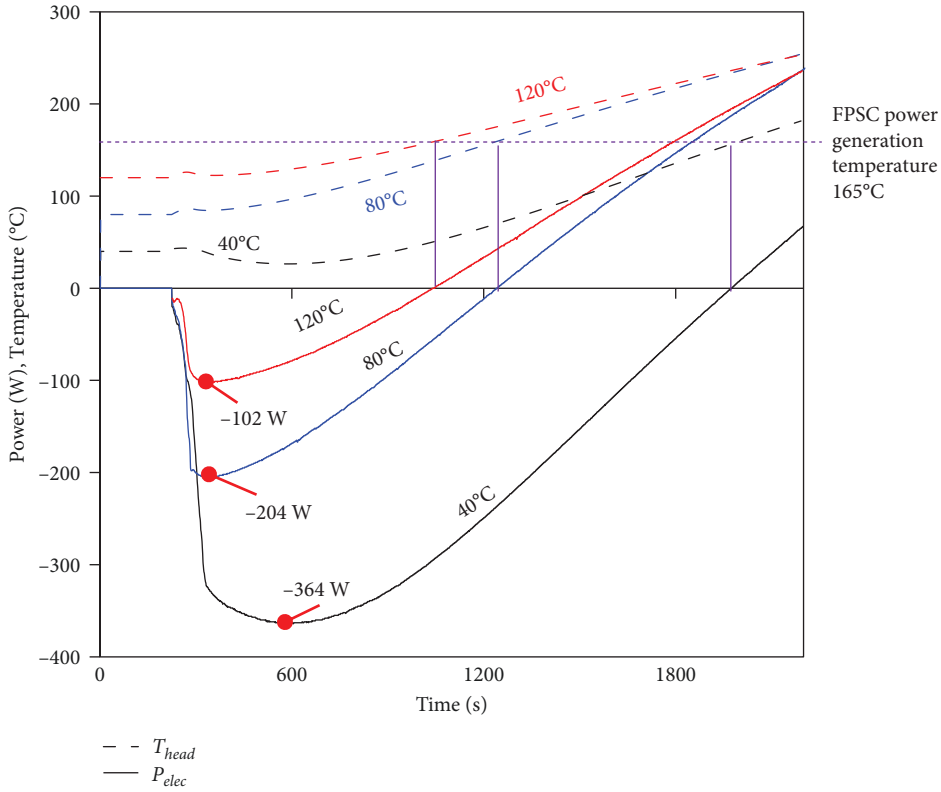


FIGURE 9: FPSC power output during startup for initial head temperatures of 40, 80, and 120°C, with negative FPSC power indicating power input from the AC bus.

TABLE 6: Measured motoring power and time for the test fpsc at the initial head temperatures of 40, 80, and 120°C.

Initial head temperature	40°C	80°C	120°C
Motoring time (sec)	1752	1018	821
Average motoring power (W)	227.1	119.2	61.2
Total amount of motoring power (Wh)	108.8	33.70	13.95

3.3. Effects of the Load Resistance of the AC Bus Circuit. Figure 13 shows the measured transient (a) FPSC power output, AC bus power output, and load power output for the test FPSC under AC bus control with load resistance values ranging from 38 to 48 Ω with an interval of 2 Ω for head temperatures from 350 to 500°C with an interval of 50°C, along with (b) the head temperature and heater input power. The control voltage and frequency are 185 V_{rms} and 50 Hz, respectively. The tuning capacitance is 50 μF. With the given head temperature maintained by adjusting the heater input power, the load resistance was decreased from 48 to 38 Ω and the head temperature was increased sequentially from 350 to 500°C.

Figure 14 shows the measured (a) FPSC power output, (b) AC bus power output, (c) load power output, (d) FPSC efficiency, and (e) power factor versus the load resistance from 38 to 48 Ω for head temperatures from 350 to 550°C. The experiment at the head temperature of 550°C was performed separately, with corresponding detailed measurements omitted here for brevity.

As a result, the FPSC power output was constant regardless of changes in the load resistance. The load power output decreased regularly as the load resistance increased at a constant control voltage, leading to decreases in the AC bus power output by the same amount. The FPSC efficiency was relatively constant regardless of the changes in the load resistance, similar to the trends of the FPSC power output. The power factor was close to 1 regardless of the load resistance.

As the head temperature increased, increases in the FPSC power output led to decreases in the AC bus power output by the same amount, while the load power output remained constant due to the fixed control voltage and the given load resistance. The FPSC efficiency and power factor increased to approximately 24% and to nearly 1, respectively.

In particular, the AC bus power output was close to 0 at the load resistance levels of 38, 42, and 48 Ω for the head temperatures of 550°C, 500°C and 450°C, respectively. This result shows that the AC bus power output, which is considered as a power loss at the system level, can be minimized around 0 by changing the load resistance for each head temperature without affecting the FPSC performance. Note that the FPSC performance varied according to the load resistance when the load control method was used, as shown in Figure 7.

3.4. Control Characteristics of the AC Bus Control Voltage. Figure 15 shows the measured transient (a) FPSC power output, AC bus power, and load power output for the test FPSC at AC bus control voltages from 160 to 210 V_{rms} with an interval of 12.5 V_{rms} at head temperatures of 350, 400, 450, and 500°C,

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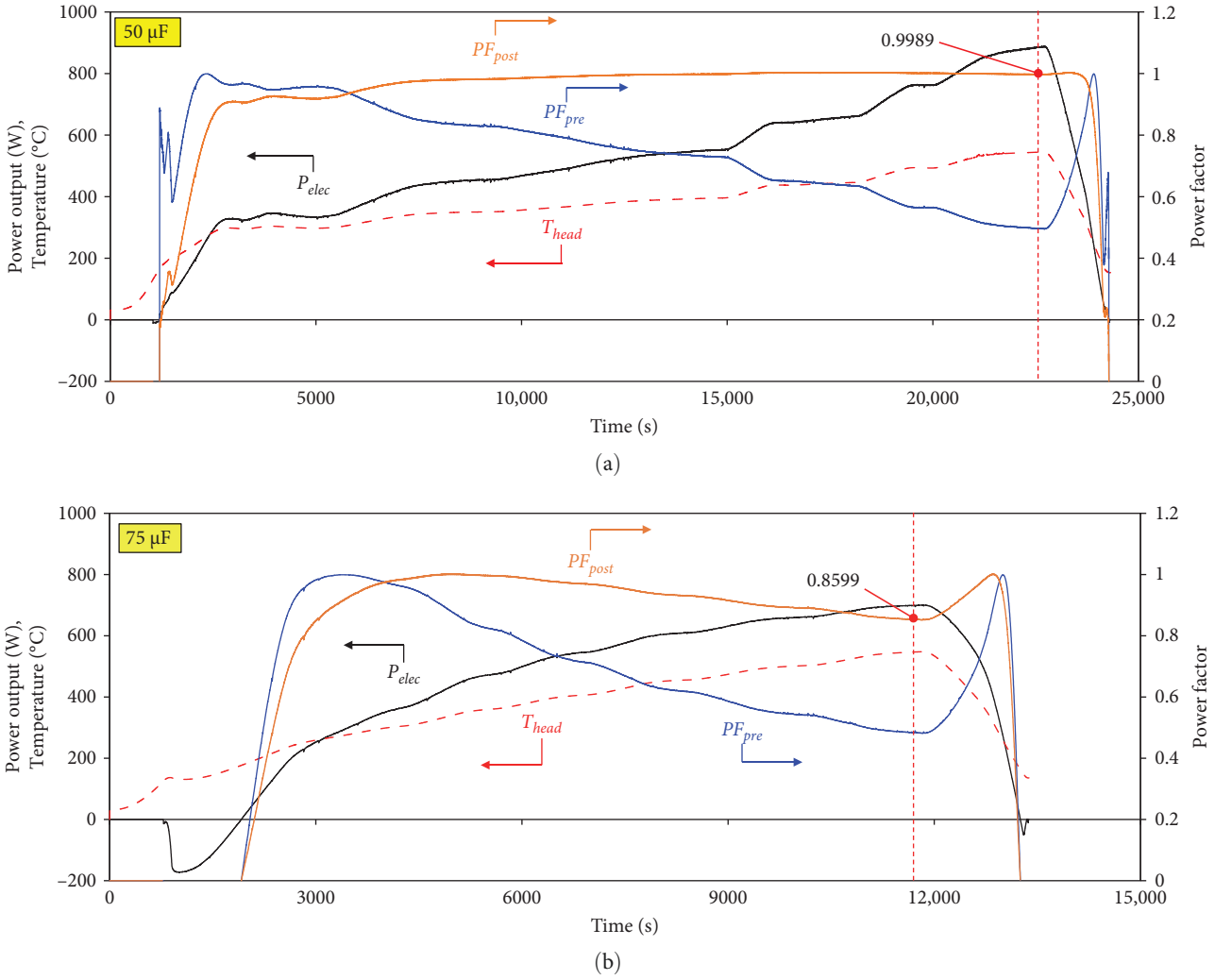


FIGURE 10: Measured transient FPSC power output levels and power factors at the pre- and post-capacitors for the test FPSC under the AC bus control scheme with tuning capacitances of (a) 50 μF and (b) 75 μF. The head temperature is increased to 550°C.

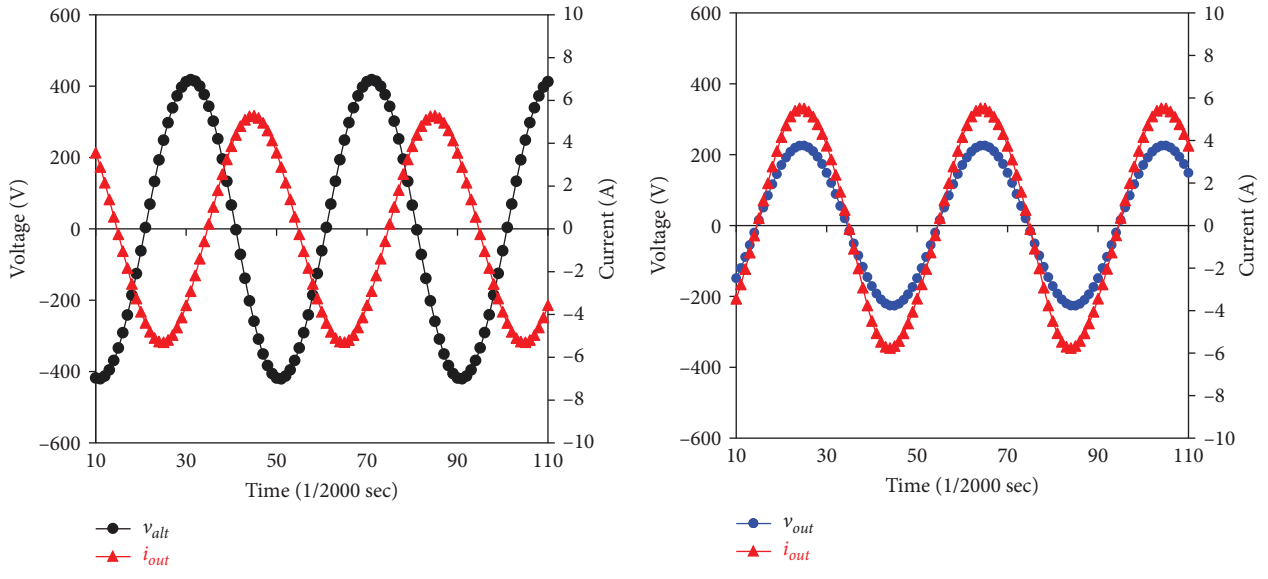


FIGURE 11: Current and voltage waveforms of (a) the pre- and (b) post-capacitor at 22,750 sec in Figure 10a.

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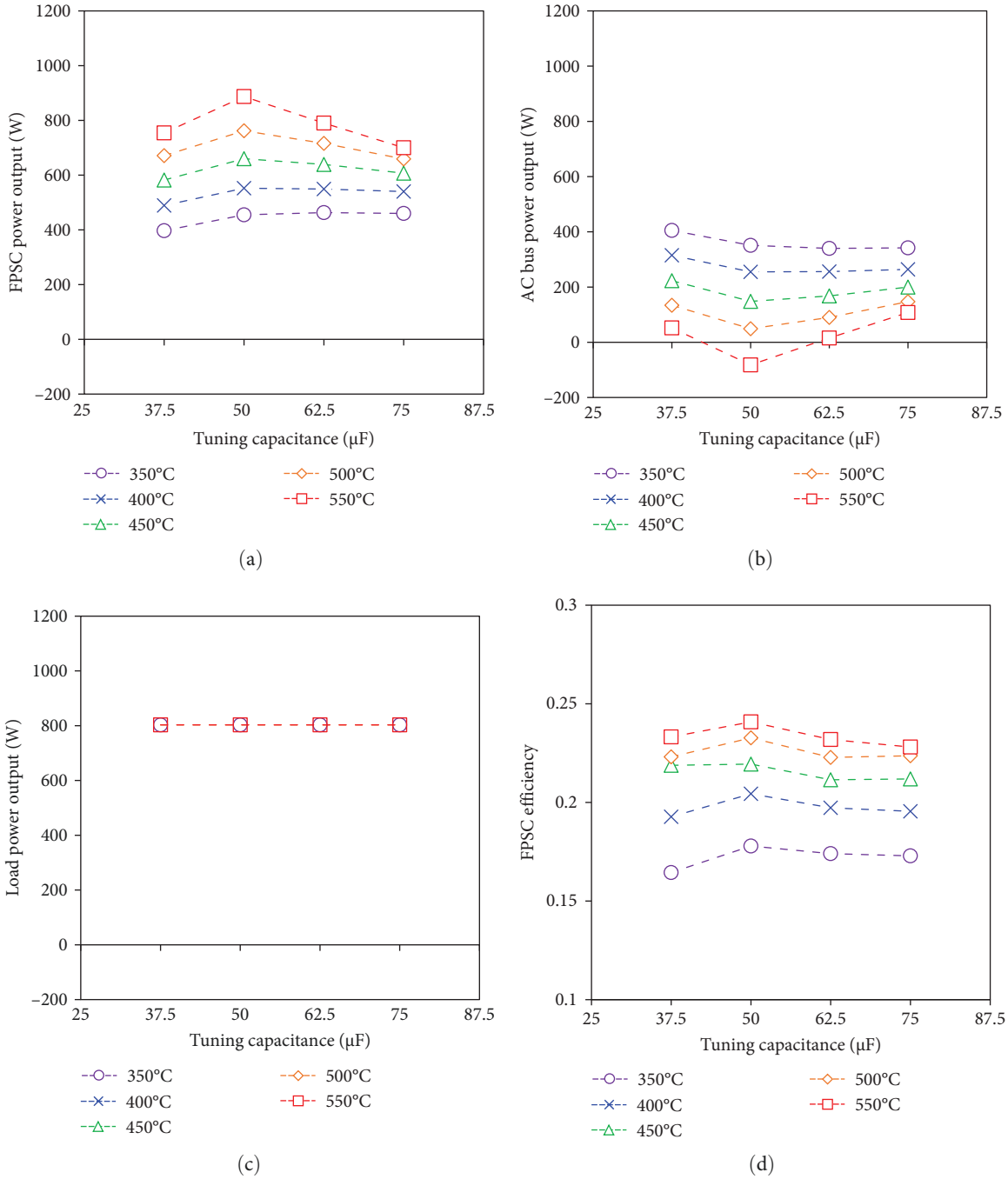
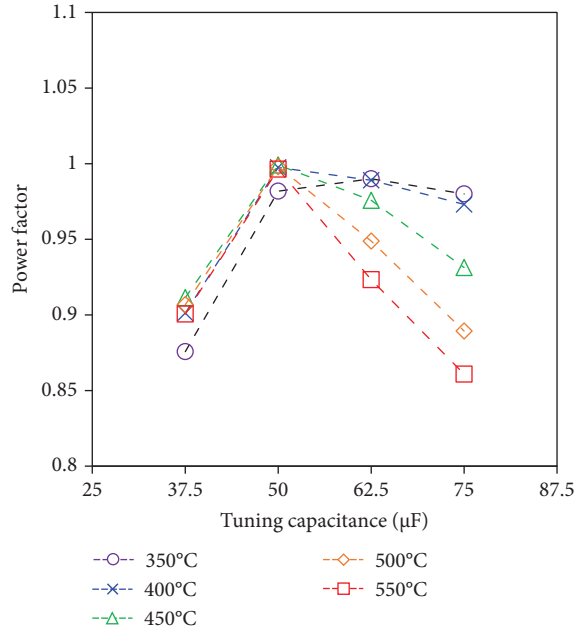


Figure 12: Continued.

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(e)

FIGURE 12: Measured (a) FPSC power output, (b) AC bus power output, (c) load power output, (d) FPSC efficiency, and (e) power factor (post-capacitor) versus the tuning capacitance of 37.5, 50, 62.5, 75 μF for head temperatures of 350, 400, 450, 500, and 550°C.

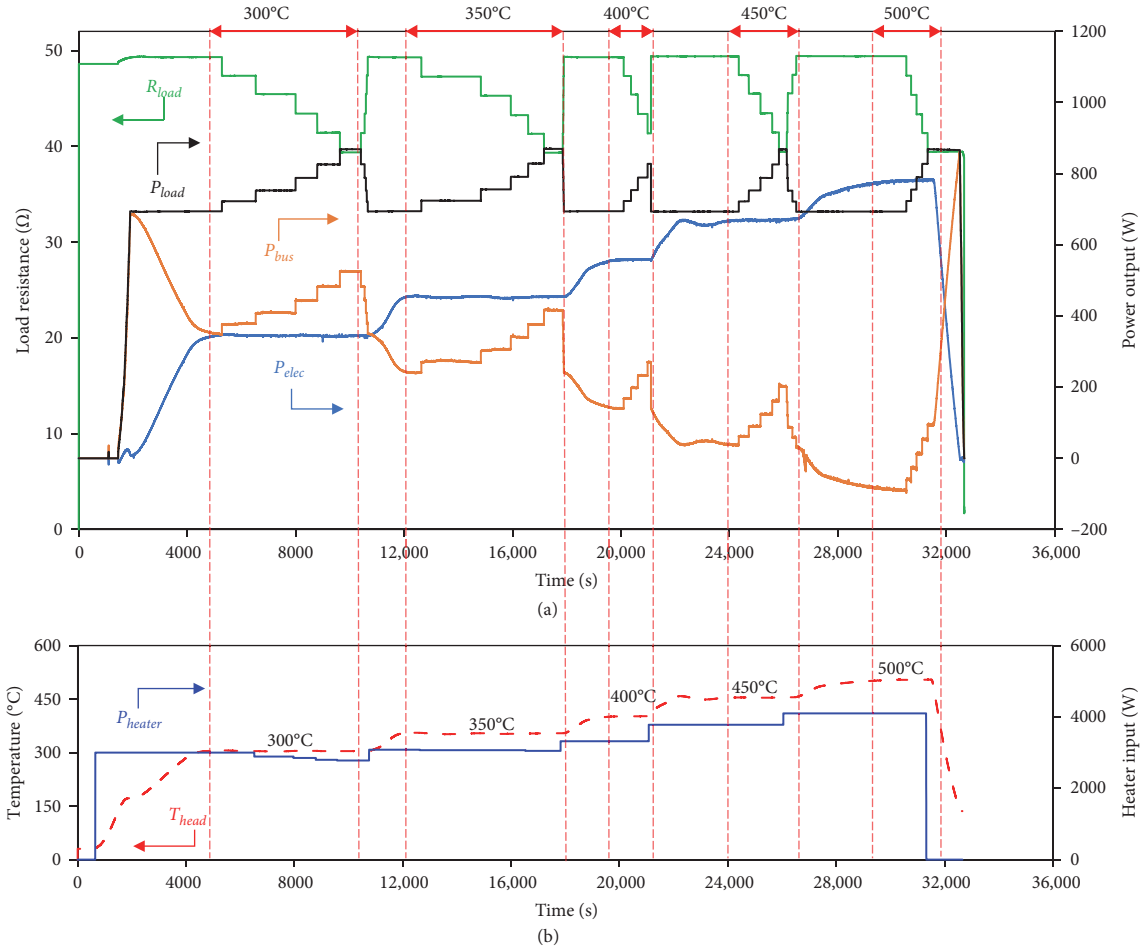


FIGURE 13: Measured transient (a) FPSC power output, AC bus power output, and load power output for the test FPSC under AC bus control with load resistance values ranging from 38 to 48 Ω with an interval of 2 Ω for head temperatures from 350 to 500°C with an interval of 50°C, along with (b) the head temperature and heater input power.

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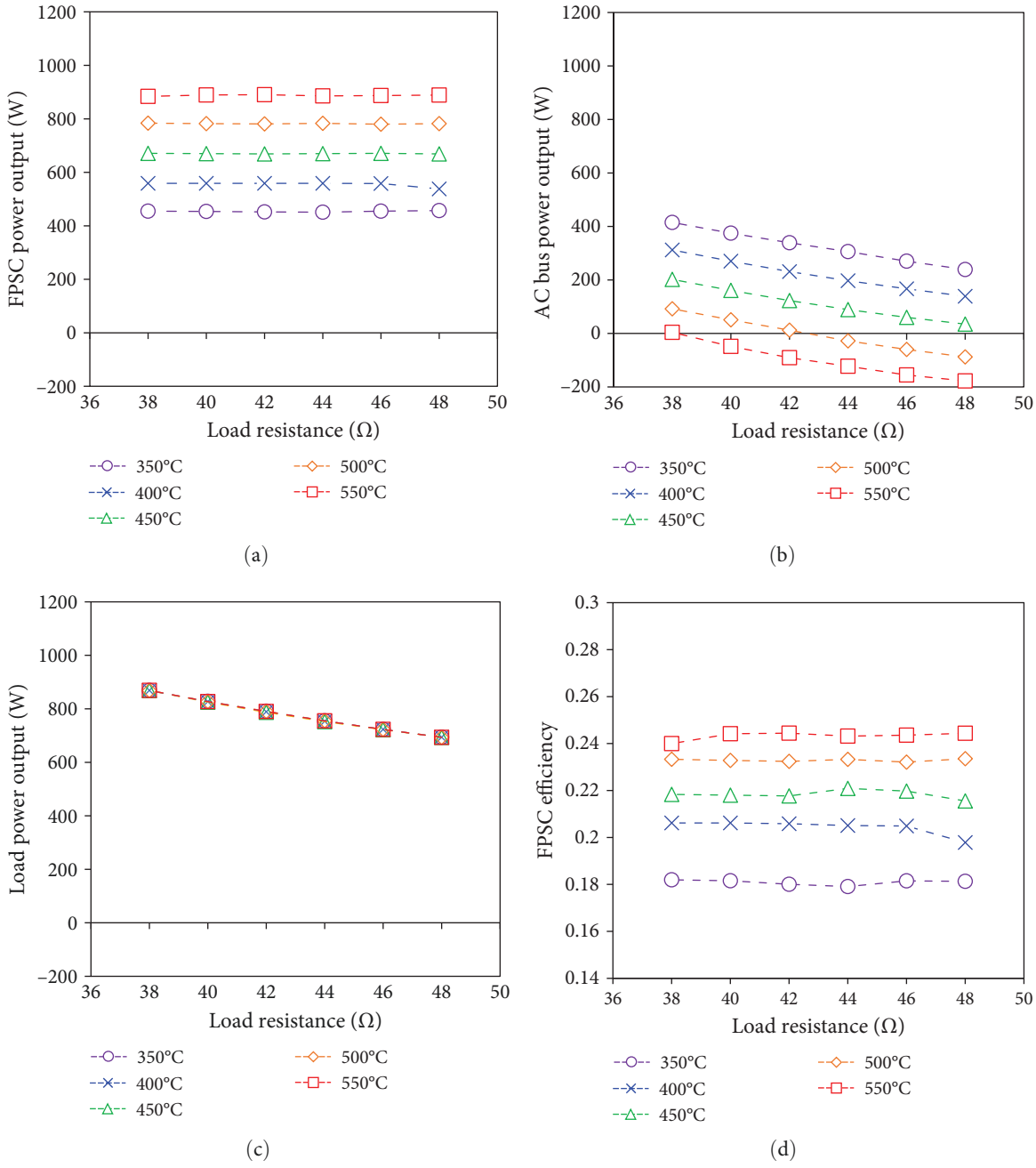
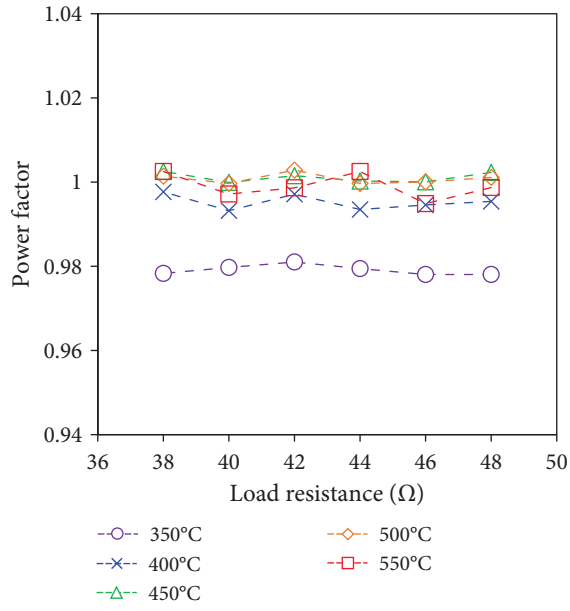


Figure 14: Continued.

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Figure 14: Measured (a) FPSC power output, (b) AC bus power output, (c) load power output, (d) FPSC efficiency, and (e) power factor versus the load resistance from 38 to 48 Ω for head temperatures from 350 to 550°C, extracted from Figure 13.

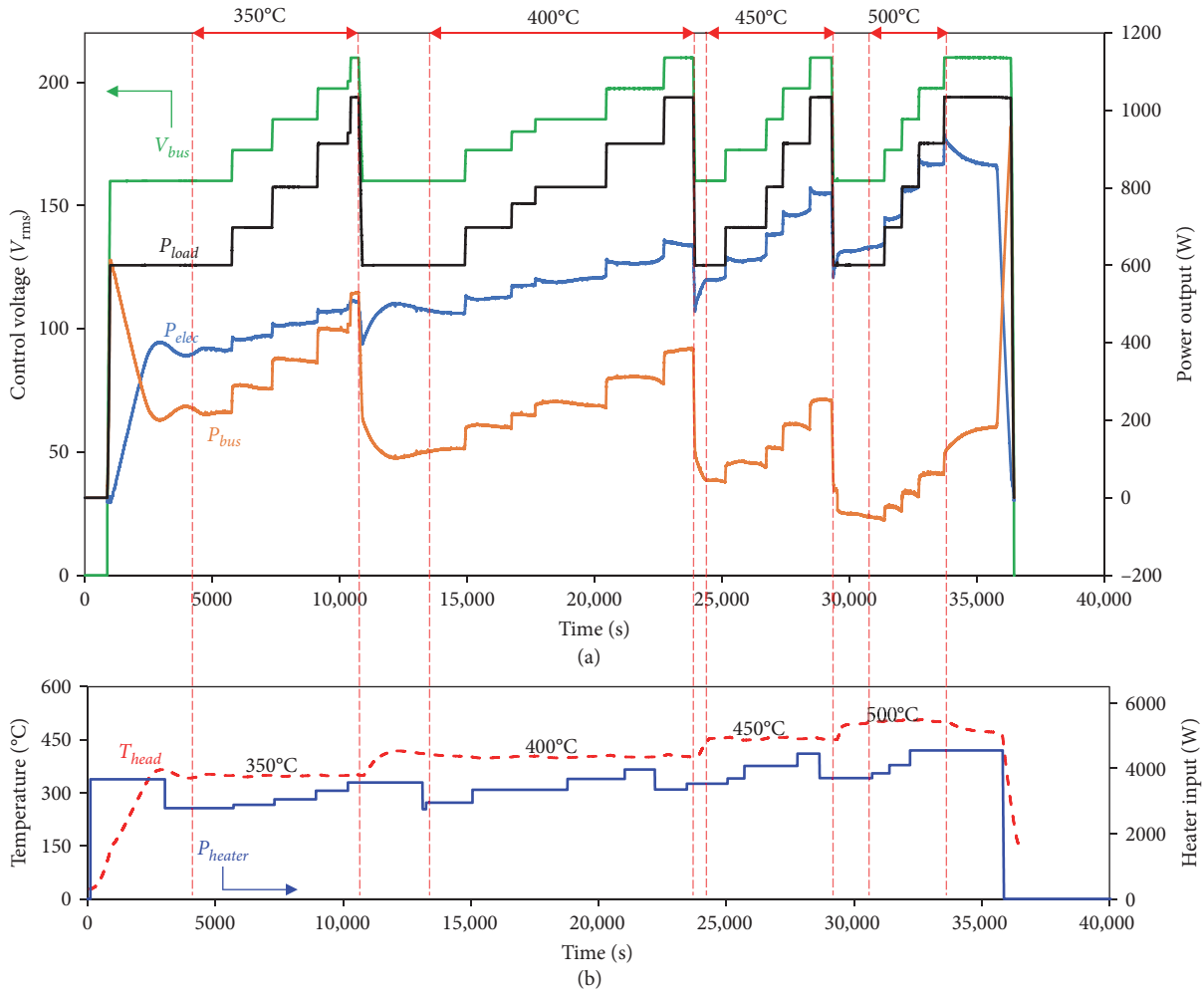


Figure 15: Measured transient (a) FPSC power output, AC bus power, and load power output for the test FPSC at AC bus control voltages from 160 to 210 V_{rms} with an interval of 12.5 V_{rms} at head temperatures of 350, 400, 450, and 500°C, along with (b) the head temperature and heater input power.

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along with (b) the head temperature and heater input power. The control frequency is 50 Hz. The tuning capacitance and load resistance are 50 μF and 42 Ω, respectively.

Figure 16 shows the measured (a) FPSC power output, (b) AC bus power output, (c) load power output, (d) FPSC efficiency, and (e) power factor versus the control voltage from 160 to 210 V_{rms} for head temperatures of 350, 400, 450, and 500°C, as extracted from Figure 15. The experiment at the head temperature of 500°C and control voltage of 210 V_{rms} is not included here because the head temperature was not maintained at 500°C due to the limited heater capacity (max. 4500 W).

As a result, the FPSC power output shows approximately linear increases with respect to the control voltage at all head temperatures. The rate of increase rises at higher head temperatures. Note that increases in the control voltage lead to increases in the PP amplitude as well as the FPSC power output [16]. Therefore, the control voltage was set below 230 V_{rms}, which is the maximum specification output voltage of the test FPSC, to avoid collisions or failures of pistons in the internal structure of the engine.

The AC bus power output and load power output also increase at higher control voltages based on the power balance, where the load power output is determined by the control voltage and the load resistance. Regarding the increase in the head temperature, the FPSC power output increases at relatively equal intervals, while the AC bus power output decreases and the load power output remains constant.

In particular, the AC bus power output becomes negative at the control voltages of 160 and 172.5 V_{rms} for the head temperature of 500°C, indicating that the FPSC power output is greater than the load power output and that part of the FPSC power output is input to the AC bus. Note that high power input to the AC bus may cause a failure of the AC bus, becoming a power loss from the system perspective. Therefore, the control voltage of approximately 185 V_{rms} is found to be optimal for a load of 42 Ω at the head temperature of 500°C, yielding FPSC power output of 800 W, nearly identical to the load power.

The FPSC efficiency tended to increase and then decrease as the control voltage was increased, reaching a maximum of 23.7% at the control voltage of 185 V_{rms}. For reference, it is known that FPSEs have an optimal piston amplitude that produces maximum efficiency [33]. The FPSC efficiency increased overall with an increase in the head temperature, as expected.

The power factor is relatively constant over the control voltages and has values close to 1 at higher head temperatures, except for 350°C. The case of 350°C is discussed in detail in the discussion section.

3.5. Control Characteristics of the AC Bus Control Frequency. Figure 17 shows the measured transient (a) FPSC power output, AC bus power output, and load power output for the test FPSC at AC bus control frequencies from 49.2 to 50.6 Hz with an interval of 0.12 Hz at a head temperature of 450°C, along with (b) the head temperature and heater input power. The control voltage is 185 V_{rms}. The tuning capacitance is 50 μF and the load resistance is 42 Ω.

The FPSC initially starts and maintains a control frequency of 50 Hz until reaching a steady state at a head temperature of 450°C, after which the control frequency is reduced to 49.2 Hz. Afterwards, the control frequency is increased stepwise to 50.6 Hz at intervals of 0.12 Hz. The heater input power was adjusted to keep the head temperature constant at 450°C.

Figure 18 shows the measured (a) FPSC power output, (b) AC bus power output, (c) load power output, (d) FPSC efficiency, and (e) power factor versus the control frequency from 49.2 to 50.6 Hz at head temperatures of 350 and 450°C as extracted from Figure 17, where the test results for 350°C are omitted for brevity.

As a result, the FPSC power output showed a tendency to increase then decrease as the control frequency was increased for both head temperatures, becoming maximized around the FPSC specification frequency of 50.0 Hz. In particular, at the head temperature of 450°C, a remarkable reduction in the FPSC power output above 50.0 Hz was noted. Because the load power output is constant regardless of the control frequency and/or head temperature, the behavior of the AC bus power output is opposite that of the FPSC power output.

The FPSC efficiency also tends to increase and then decrease according to the control frequency, while the maximum efficiency is shown more clearly at the control frequencies of 49.84 and 50.08 Hz for the head temperatures of 350 and 450°C, respectively. Similar to the FPSC efficiency, the power factor increases to 1 and then decreases, with the control frequency with the maximum power factor rising as the head temperature increases.

Note that the engine's natural frequency is found to rise slightly due to the increase in the internal pressure and thus the pressure stiffness of the engine at higher head temperatures [16]. This phenomenon is also shown in Figure 7, where the FPSC performance measured with no active control shows increases in the operating frequency from 49.5 to 50.02 Hz as the head temperature increases from close to 300 to 450°C. Interestingly, the power factor shows sensitive variations for small differences of less than 1 Hz between the system natural frequency and the control frequency. This is discussed in detail in the discussion section.

4. Discussion

4.1. Influence of the Alternator Inductance on the Control Characteristics. In the AC bus control experiments, the power factor decreased significantly at the head temperature of 350°C, as shown in Figures 14e and 16e, and also with minute changes in the control frequency of less than 1 Hz near 50 Hz, as shown in Figure 18e. Moreover, the power factor is clearly shown to increase when the head temperature increases from 350 to 550°C, approaching 1 in the transient graph in Figure 10a.

To analyze these phenomena, a phasor diagram analysis was conducted using a voltage equation in the complex form converted from Equation (1). This is expressed as follows:

$$V_{bus} + V_{emf}e^{j\theta_i} - \frac{1}{j\omega C}I_{out}e^{j\theta_i} - R_{alt}I_{out}e^{j\theta_i} - j\omega L_{alt}I_{out}e^{j\theta_i} = 0, \tag{3}$$

where I_{out} and θ_i are correspondingly the RMS amplitude and phase angle of the FPSC output current i_{out} with respect

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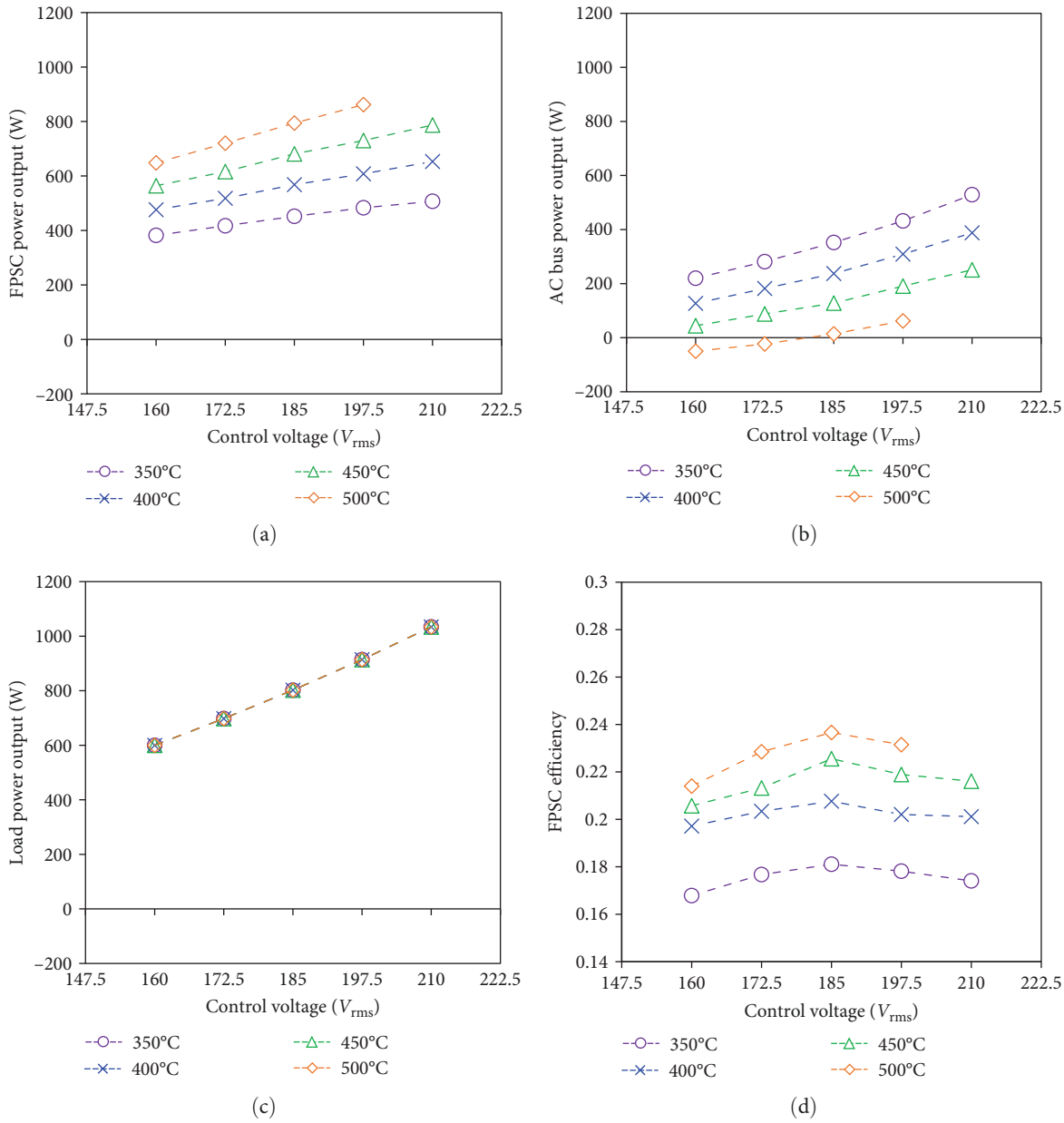
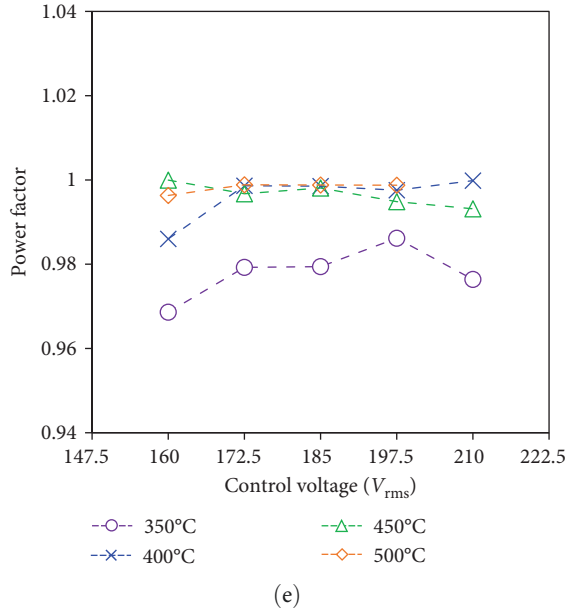


FIGURE 16: Continued.

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FIGURE 16: Measured (a) FPSC power output, (b) AC bus power output, (c) load power output, (d) FPSC efficiency, and (e) power factor versus the control voltage from 160 to 210 V_{rms} for head temperatures of 350, 400, 450 and 500°C, extracted from Figure 15.

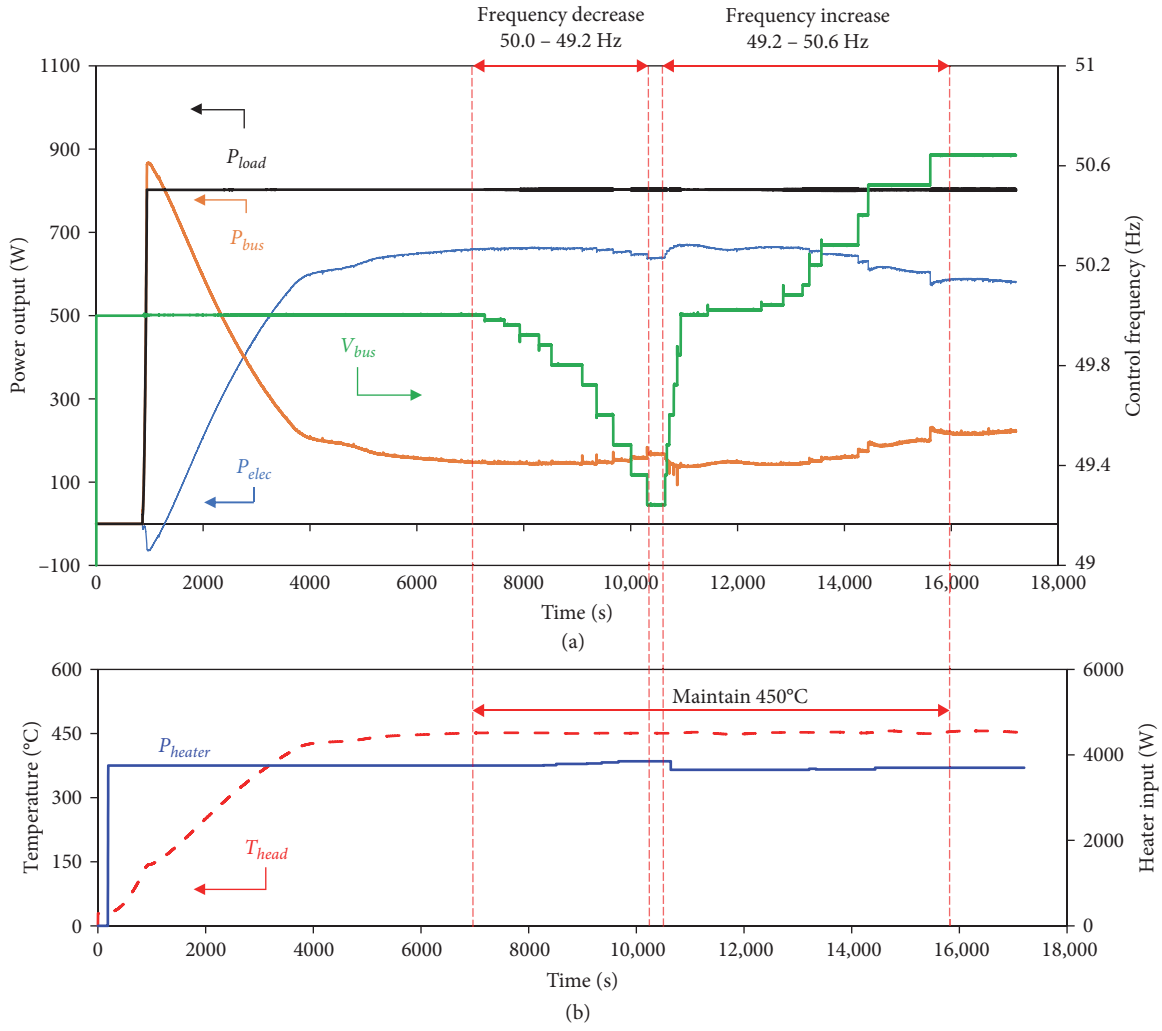


FIGURE 17: Measured transient (a) FPSC power output, AC bus power output, and load power output for the test FPSC at AC bus control frequencies from 49.2 to 50.6 Hz with an interval of 0.12 Hz at a head temperature of 450°C, along with (b) the head temperature and heater input power.

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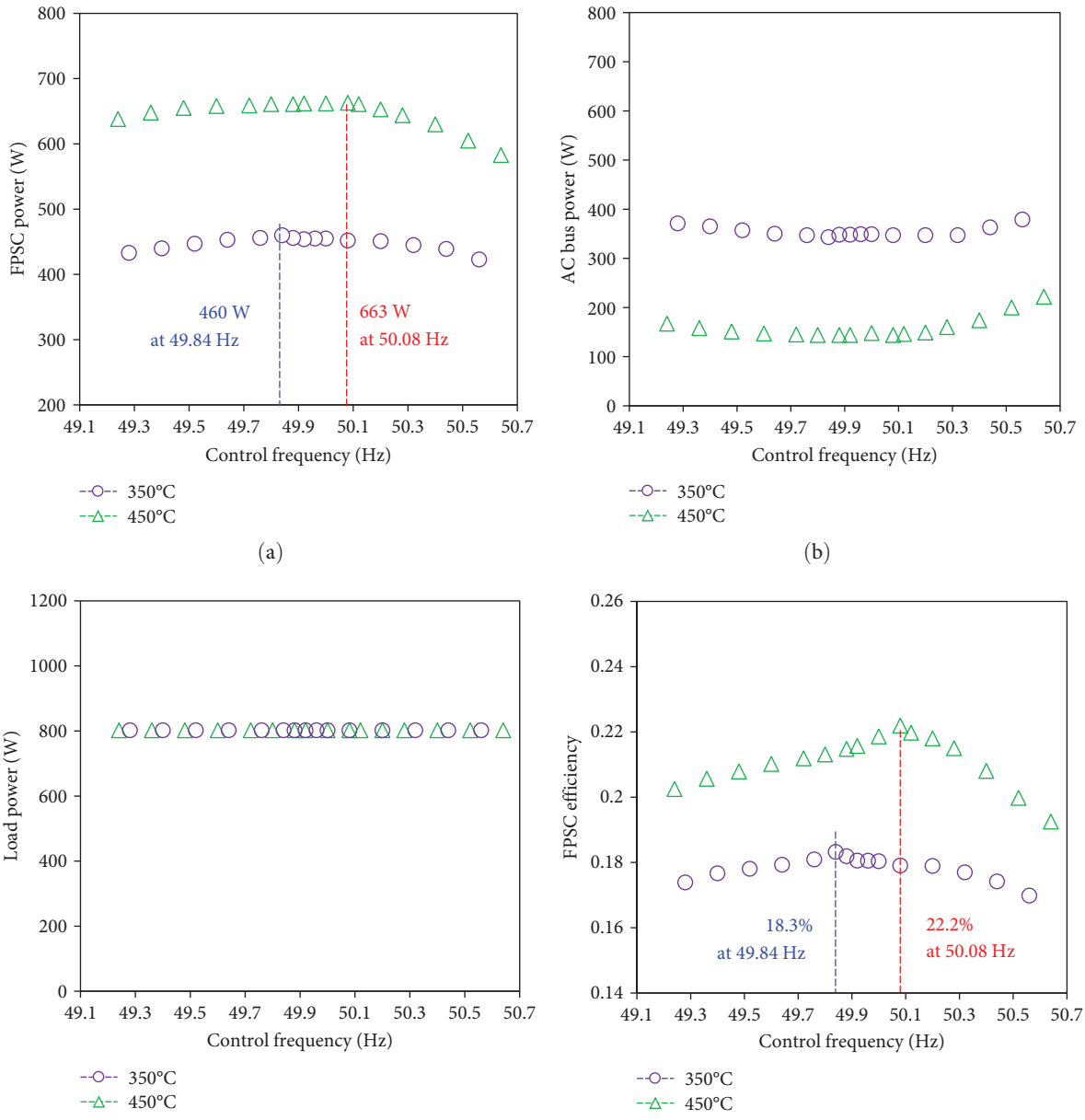
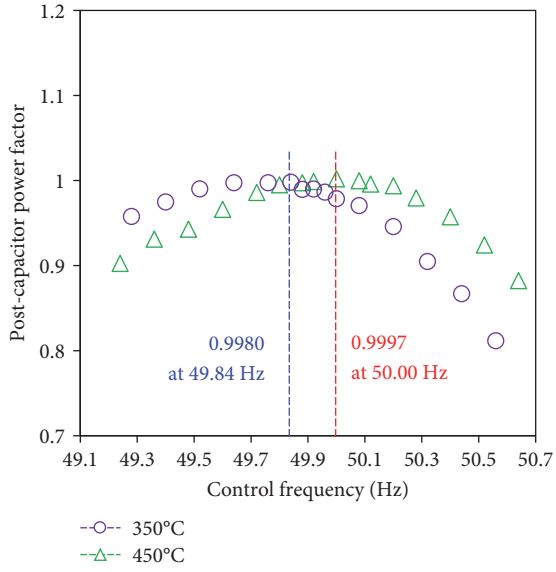


Figure 18: Continued.

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FIGURE 18: Measured (a) FPSC power output, (b) AC bus power output, (c) load power output, (d) FPSC efficiency, and (e) power factor versus control frequency from 49.2 to 50.6 Hz at head temperatures of 350°C, and 450°C, extracted from Figure 17.

to the phase of the AC bus voltage v_{bus}. j refers to $\sqrt{-1}$. The negative sign for the last three terms represents the voltage drop.

The induced voltage V_{emf} is calculated from $P_{\text{elec}} = V_{\text{emf}} I_{\text{out}} \cos\varphi_{\text{emf}} - {I_{\text{out}}}^2 R_{\text{alt}}$ [34], where the power factor $\cos\varphi_{\text{emf}}$ is 1 because the induced voltage and output current of the linear alternator are assumed to be in phase during mechanical resonance operations of FPSCs [35]. Note that under mechanical resonance, the mover velocity and force become in-phase, with each having the same phase with the induced voltage and output current, respectively, with proportionality of the motor constant of the alternator.

Finally, the alternator inductance can be calculated by substituting the measured FPSC output power, output current and the AC bus control voltage into the voltage equation.

Figure 19 shows (a) the calculated inductance and induced voltage, and the measured output current; (b) the calculated inductive and capacitive reactance, and the power factor; and (c) a voltage phasor diagram for head temperatures from 350 to 500°C, corresponding to the test results in Figure 15. The control voltage and frequency are 185 V_{rms} and 50 Hz, respectively. The tuning capacitance and load resistance are likewise 50 μF and 42 Ω. The inductive reactance $X_{L.\text{alt}}$ and capacitive reactance $X_{C.t}$ are correspondingly defined as $2\pi f L_{\text{alt}}$ and $1/2\pi f C_t$.

When the head temperature increases, the induced voltage remains nearly constant with slight increases, while the output current increases significantly by approximately 70%. Therefore, the FPSC power output increases mainly due to the increase in the output current. Because the magnetic permeability μ increases with the electric current before magnetic saturation [36], the inductance increases by about 40% from 152.6 to 210.3 mH. In particular, the inductive reactance at the head temperature of 350°C is noticeably lower than the capacitive reactance, while the conductive reactance is constant. As a result, the power factor is clearly reduced at 350°C.

Note that the inductance of the alternator is a variable value that is defined as $L_{\text{alt}} = \mu {N_{\text{coil}}}^2 A_{\text{coil}} / l_{\text{coil}}$ [23], where the number of turns of the coil N_{coil}, the cross-sectional area of the coil A_{coil}, and the length of the coil l_{coil} are fixed values, while the magnetic permeability of the core, which is defined as $\mu = \text{B/H}$, is a variable that changes according to the magnetic field strength H and magnetic flux density B [37]. The magnetic permeability is known to have a nonlinearity that increases according to the magnetic field strength and then decreases as magnetic saturation occurs. The magnetic field strength is proportional to the current. Therefore, an increase in the current results in an increase in the alternator inductance before magnetic saturation.

In the phasor diagram, $v_{L.\text{alt}}$ is also smaller than $v_{C.t}$, resulting in the phase of v_{emf} being delayed by about 11.6° with respect to the output voltage v_{out} of the FPSC and tuning capacitor, which is defined as the vector sum of v_{emf}, $v_{R.\text{alt}}$, $v_{L.\text{alt}}$, and $v_{C.t}$, thus having the same magnitude but the opposite direction relative to v_{bus}.

Note that the phases of $v_{L.\text{alt}}$ and $v_{C.t}$ are perpendicular to v_{emf}, delayed and leading by 90°, respectively, while $v_{R.\text{alt}}$ is out-of-phase with a relatively small magnitude. The phase of the output current is identical to that of v_{emf}.

As the head temperature increases, $v_{L.\text{alt}}$ increases more than $v_{C.t}$, and the phase difference of v_{emf} from v_{out} decreases and approaches 0 at the head temperature of 450°C. The entire phasor diagram is shown to rotate minutely in the counterclockwise direction.

Consequently, the head temperature, which is an operating condition of FPSCs unrelated to the AC bus control parameters, is found to have a significant influence on the control characteristics of the power factor by changing the alternator inductance with the output current.

Figure 20 shows (a) the calculated inductance and induced voltage, as well as the measured output current; (b) the calculated inductive and capacitive reactance, and the power factor;

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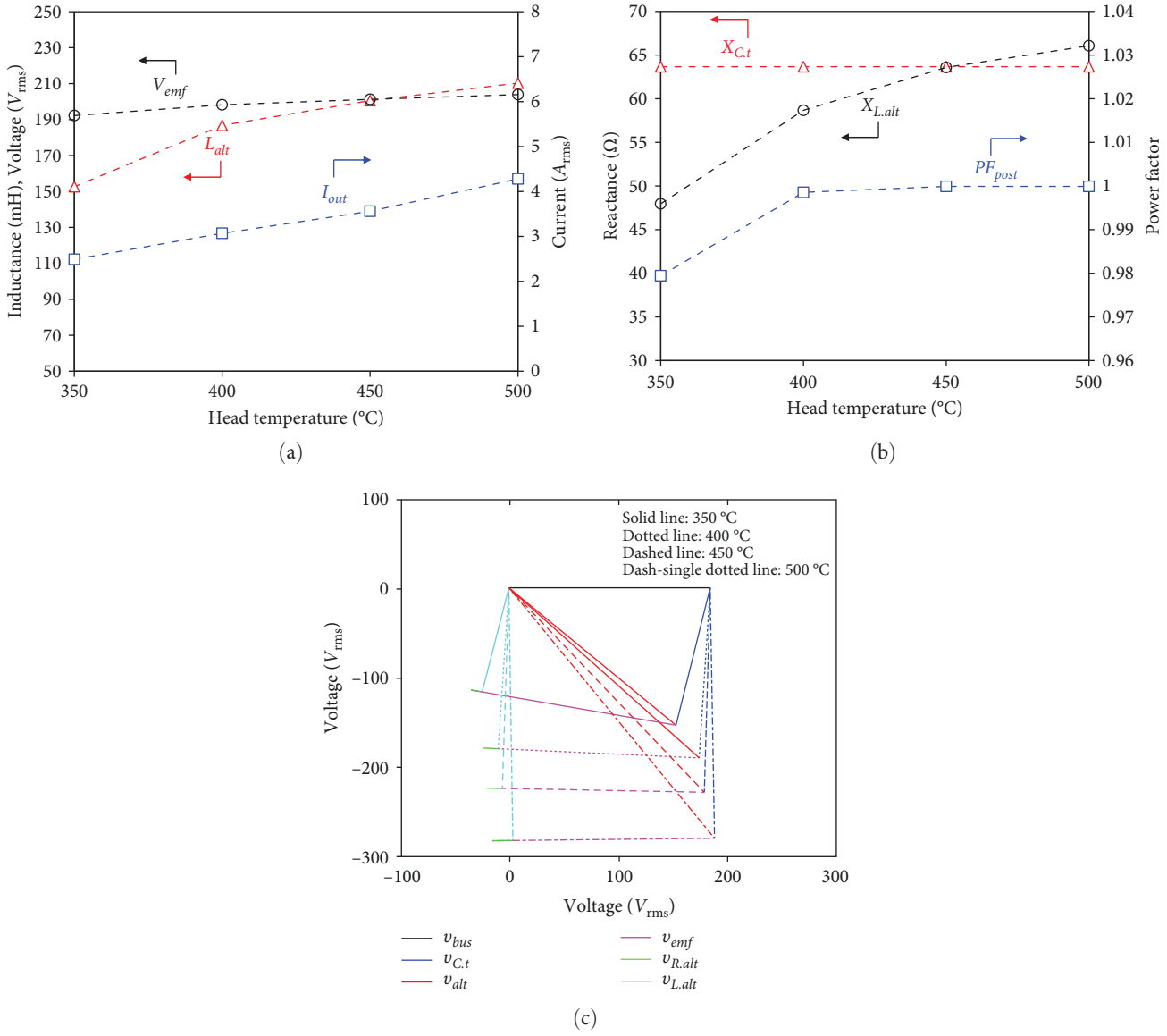


FIGURE 19: (a) calculated inductance and induced voltage, and measured output current; (b) calculated inductive and capacitive reactance, and power factor; and (c) voltage phasor diagram for head temperatures from 350 to 500°C, corresponding to the test results in Figure 15.

and (c) a voltage phasor diagram for control frequencies from 49.24 to 50.64 Hz, corresponding to the test results shown in Figure 17.

As the control frequency increases, the inductance decreases by approximately 57% from 274.2 to 119.4 mH. At the same time, the measured output current decreases, causing a reduction in the magnetic field strength [36]; the magnetic field direction changes more rapidly due to the increased frequency, causing the magnetic flux density to decrease [38]. As a result, the inductive reactance decreases and becomes equal to the capacitive reactance at the control frequency of 50 Hz, resulting in the maximum power factor.

Regarding the voltage phasor diagram, the inductance voltage is greater than the capacitance voltage at the lower control frequency of 49.24 Hz, yielding the phase of v_{emf} ahead of v_{out} by 25.5°. As the control frequency increases, the inductance voltage decreases, the phasor diagram rotates clockwise, and the phase angle and thus the power factor change significantly. As a result, the phase angle of v_{emf} becomes zero near the control frequency of 50 Hz, and the magnitude is maximized.

5. Conclusion

This study presents an experimental assessment of the performance measurements and control characteristics of a 1 kW class commercial FPSC with an AC bus control scheme. An FPSC performance test rig is constructed, and an AC bus controller is developed. Comprehensive experimental parameters include the AC bus circuit parameters of the tuning capacitance and the load resistance along with the AC bus control

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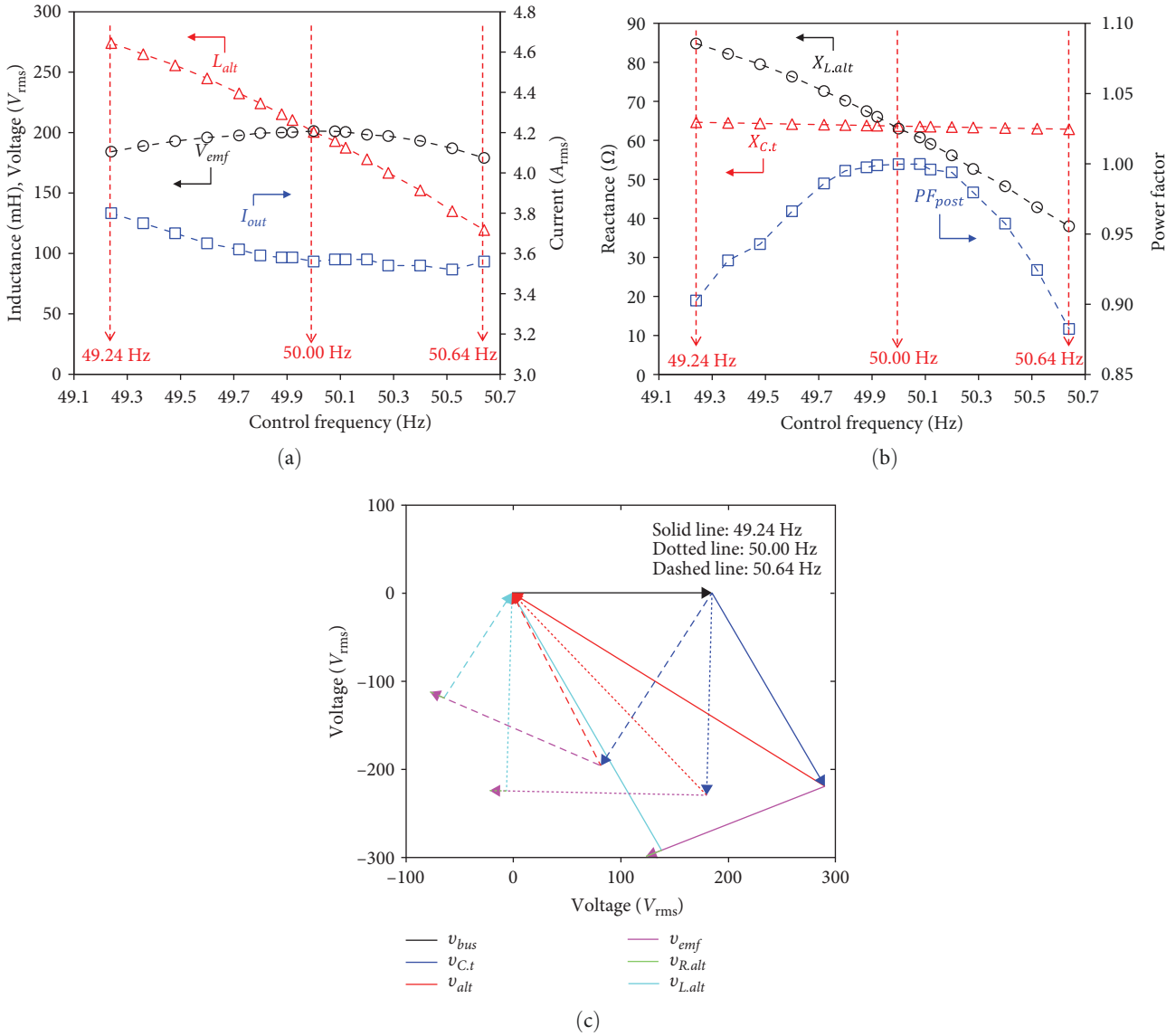


FIGURE 20: (a) calculated inductance and induced voltage, and measured output current; (b) calculated inductive and capacitive reactance, and power factor; and (c) voltage phasor diagram for control frequencies from 49.24 to 50.64 Hz, corresponding to the test results in Figure 17.

parameters of the control voltage and control frequency. The experimental design of the AC bus parameters was determined through a series of preliminary tests under load resistance control without external active controls. Through this, the following conclusions were obtained.

5.1. Tuning Capacitance. The FPSC power and efficiency are maximized at the optimal tuning capacitance due to a higher power factor, as expected. The tuning capacitance matches the resonant frequency of the AC bus circuit to the natural operating frequency of the test FPSC, leading to a phase difference of the output current voltage near 0 and a power factor approaching 1.

However, the effects of correcting the power factor with a fixed tuning capacitance were unequal over the head temperatures and FPSC performances here due to changes in the alternator inductance. This fact implies that the tuning capacitance must be optimized under off-design conditions as well.

5.2. Load Resistance. Changes in the load resistance affected only the AC bus output and the load power output based on the power balance of the AC bus circuit. They did not influence the FPSC power output or efficiency. This reveals that there is no direct relationship between the load resistance and FPSC performance under an AC bus control scheme.

This fact implies that the load resistance of the AC bus control circuit can be adjusted to minimize the AC bus power output without affecting the FPSC performance. In another sense, the AC bus output can be used to compensate for fluctuating load demands by supplementing the FPSC output, which is fixed according to the control voltage.

5.3. Control Voltage. The FPSC power output showed approximately linear increases as the control voltage was increased. The load power output was determined by the control voltage with the given load resistance. The power factor is relatively

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constant over the control voltages, not showing a clear correlation. On the other hand, the FPSC efficiency tended to increase and then decrease, reaching its maximum in the middle of the control voltage range.

The control voltage can freely control the piston amplitude to produce a specific output, as the piston amplitude is directly determined by the control voltage with the given motor constant and operating frequency.

5.4. Control Frequency. The FPSC power output and efficiency were maximized at an optimal control frequency near the FPSC-specified operating frequency of 50 Hz. Furthermore, the optimal control frequency changed as the head temperature increased, as shown with 49.84 and 50.08 Hz at the head temperatures of 350°C and 450°C, respectively. This likely occurred because the FPSC natural operating frequency was increased due to the higher FPSC pressure stiffness at the higher head temperature.

The power factor showed a trend similar to that of the FPSC performance, varying from 0.8 to 1 even with small frequency changes of less than 1 Hz. From the AC bus circuit analysis, it was found that the alternator inductance was reduced considerably over the control frequencies, leading to a power factor of 1 near the optimal frequency.

The power factor was found to be strongly related to the FPSC performance, considerably influenced by the control frequency. Furthermore, the range of control frequency with higher power factors is very narrow at less than 1 Hz. Therefore, the control frequency must be adjusted accurately to the optimal frequency according to the changes in the head temperature to maximize the FPSC performance.

Hence, the AC bus control scheme is shown to provide direct control of the piston motion with a simple circuit including a tuning capacitor and a power supply. Significant changes in the alternator inductance depending on the head temperature and control frequency are found to require a tuning capacitor for the power factor correction to be adjusted precisely for maximum performance of a FPSC, especially under off-design conditions. This work is expected be helpful to those who design virtual capacitor controllers while also providing a simple and accurate control method for FPSCs in laboratory experiments.

Nomenclature

Symbols

A_{heater}: Area of the outer surface of the heater (m^2)
A_{coil}: Cross-sectional area of the coil (m^2)
B: Magnetic flux density (T)
C_t: Tuning capacitance (F)
f: Operating frequency (Hz)
H: Magnetic field strength (A/m)
$\overline{h}$: Mean convective heat transfer coefficient (W/m^2)
I: RMS amplitude of the current (A_{rms})
i: Current (A)
K_e: Motor constant (N/A)
L_{alt}: Coil inductance of the linear alternator (H)
l_{coil}: Length of the coil (m)
N: Number of direct measurement parameters (−)
N_{coil}: Number of turns of the coil (−)
P: Power output (W)
P_{elec}: FPSC power output (W)
P_{heater}: Heater input power (W)
PF: Power factor (−)
$\dot{Q}_{\text{loss}}$: Heater heat loss (W)
R: Resistance (Ω)
$T_{c.\text{in}}$: Coolant inlet temperature (°C)
$T_{c.\text{out}}$: Coolant outlet temperature (°C)
T_{head}: Head temperature (°C)
T_{heater}: Heater inside temperature (°C)
T_{wall}: Temperature of the outer wall of the heater (°C)
T_∞: Ambient room temperature (°C)
V: RMS amplitude of voltage (V_{rms})
v: Voltage (V)
X: Reactance (Ω)
X_p: Amplitude of the mover (m)
x_i: Direct measurement parameter (−)
y: Indirect measurement parameter (−)

Greek Symbols

δx_i: Uncertainty of the direct measurement parameter (−)
δy: Uncertainty of the indirect measurement parameter (−)
η: FPSC efficiency (−)
η_{heater}: Heater efficiency (−)
θ_i: Phase angle of the alternator current (°)
μ: Magnetic permeability of the core (H/m)
φ: Phase angle between the voltage and current (°)

Subscripts

alt: Linear alternator
bus: AC bus
$C.t$: Capacitance
emf: Induced electromotive force
L.alt: Coil inductance of the linear alternator
load: Load resistance
out: Output
post: Post-capacitor
pre: Pre-capacitor
R.alt: Resistance of the alternator coil

Acronyms

DP: Displacer piston
FPSC: Free piston Stirling converter
FPSE: Free piston Stirling engine
GRC: Glenn research center
NASA: National aeronautics and space administration
PP: Power piston
PWM: Pulse width modulation
RMS: Root mean square.

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Appendix

Calculation of the Furnace Heater Loss. The furnace heater supplies heat to the engine head in the form of internal convection and radiation. However, thermal loss occurs in the form of a natural convective heat transfer between the outer wall of the heater and the atmosphere, which is calculated via $\dot{Q}_{\text{loss}} = \overline{h} A_{\text{heater}} (T_{\text{wall}} - T_{\infty})$, where $\overline{h}$ is the mean convective heat transfer coefficient, A_{heater} is the area of the outer surface of the heater, T_{wall} is the temperature of the outer wall surface of the heater, and T_{∞} is the ambient room temperature [39].

Table A1 shows the heat loss of the furnace heater calculated based on natural convection on the outer surfaces with the measured outer surfaces and room temperatures. The head temperature was maintained at 500°C. The heat transfer coefficients on the top, side, and bottom surfaces of the heater are defined differently depending on the direction of the outer surface.

As a result, the total heat loss was calculated and found to be 724.6 W for the heater input of 4100 W, yielding a heater loss of 17.7%. In addition, the heat loss on the side surface was greater than those of the other top and bottom surfaces. Based on this result, we assumed a heater efficiency rate of 82.3%.

Table A1: Heat losses of the furnace heater calculated based on natural convection on the outer surfaces at the head temperature of 500°C.

Variable	Outer surfaces			Unit
	Top	Side	Bottom	
T_{wall}	100	140	60	°C
T_{∞}		20		
A_{heater}	0.193	0.608	0.715	m^2
$\overline{h}$	7.1218	8.1088	3.2710	W/m^2 °C
$\dot{Q}_{\text{loss}}$	110.0	591.6	23	
Total $\dot{Q}_{\text{loss}}$		724.6		W
Heater input power		4100		
Heater efficiency		82.3%		%

Data Availability Statement

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

Jong-Su Park: Data curation, Investigation, Writing- Original draft preparation. **Dong-Jun Kim:** Software, Investigation, Writing- Reviewing and Editing. **Kyuho Sim:** Conceptualization, Methodology, Supervision, Writing- Reviewing and Editing. Jong-Su Park and Dong-Jun Kim contributed equally to this work.

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